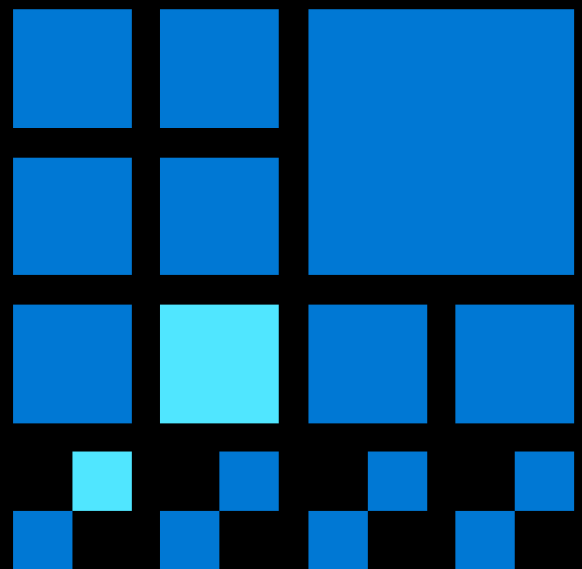


Microsoft SQL Server 2022

Licensing guide



Contents

Overview	3
SQL Server editions	4
SQL Server and Software Assurance	7
How SQL Server licenses are sold	9
Server and Cloud Enrolment	
SQL Server licensing models	11
Core-based licensing	
Server+CAL licensing	
Licensing SQL Server components	14
Licensing SQL Server in a virtualized environment	15
Licensing individual virtual machines	
Licensing for maximum virtualization	
Licensing SQL Server in containers	19
Licensing individual containers	
Licensing containers for maximum density	
Advanced licensing scenarios and detailed examples	23
Licensing SQL Server for high availability and disaster recovery	
SQL Server Azure Hybrid Benefit	
Licensing SQL Server for application mobility	
Licensing SQL Server for non-production use	
Licensing SQL Server in a multiplexed application environment	
Additional product information	35
SQL Server migration options for Software Assurance customers	
Licensing SQL Server for the Analytics Platform System	
Additional product licensing resources	

Overview

This licensing guide is for people who want to gain a basic understanding of how Microsoft® SQL Server® 2022 database software is licensed through Microsoft Volume Licensing programs. This guide does not supersede or replace any of the legal documentation covering SQL Server 2022 use rights. Specific product license terms are defined in the product Software License Terms—or in the case of Microsoft Volume Licensing—in the Microsoft Volume Licensing agreement under which the software was acquired and/or the Microsoft Volume Licensing Product Terms. This licensing guide is not a legal use rights document. Program specifications and business rules are subject to change.

SQL Server 2022 editions

Overview

SQL Server 2022 is offered in two main commercial editions to accommodate the unique feature, performance and price requirements of organizations and individuals:

- **Enterprise Edition** is ideal for applications requiring mission critical in-memory performance, security and high availability.
- **Standard Edition** delivers fully featured database capabilities for mid-tier applications and data marts.

The editions are offered in a straightforward, tiered model that creates greater consistency across the product editions, features and licensing. Enterprise Edition includes all the capabilities available in SQL Server

Through a consistent programming interface across all editions, it is easier than ever for developers and partners to build and upgrade applications that take advantage of advanced performance, security, and data mart or data warehouse features – regardless of scale, from Express to Enterprise.

SQL Server Standard Edition sets the bar for rich programming capabilities, security innovations, and fast performance for mid-tier applications and data marts. SQL Server Standard Edition can easily be upgraded to Enterprise Edition as workloads scale or for mission critical workloads without requiring an application re-write. Enterprise Edition continues to deliver the highest levels of mission critical scalability, availability, and performance as well as maximum virtualization rights with Software Assurance.

Note: SQL Server 2022 was released on November 16, 2022 for EA/EAS, MPSA, Select+, and Open Value customers. Any customer that acquired SQL Server licenses through one of those designated channels on November 16, 2022 or later is entitled to run SQL Server 2022 software with those licenses once the media is made available to them between November 2022 and January 2023. SQL Server 2022 will be released in CSP in January 2023.

SQL Server 2022 new features		Express	Standard	Enterprise
Compute and storage	Maximum number of cores	4	24	OS Max
	Maximum memory utilized per instance	1410 MB	128 GB	OS Max
	Maximum relational database size	10 GB	524 PB	524 PB
	Azure Synapse Link for SQL		×	×
Azure-enabled	Ledger	×	×	×
	Link feature for Azure SQL Managed Instance			×
	Link feature for Azure SQL Managed Instance (basic availability groups)		×	
	Microsoft Purview policies	×	×	×
Choice of language and platform	JSON T-SQL Enhancements	×	×	×
	Modern T-SQL surface area	×	×	×
	Query Store by default for new databases	×	×	×
	Query Store: Read-replica support			×
Industry-leading performance and availability	Built-in query intelligence: Parameter Sensitive Plan (PSP) Optimization, Optimized plan forcing	×	×	×
	Built-in query intelligence: CE Feedback, DOP Feedback, Memory Grant Feedback			×

	Contained Availability Group			×
	Buffer Pool Parallel Scan		×	×
			(limited to 2 threads, over 64GB Mem only)	
	AVX 512 extension for batch mode			×
	Intel QAT Enhancements for Backup		×	×
			(SW only)	(HW + SW)
	Timeseries support	×	×	×
Secure and reliable	Always Encrypted with secure enclaves - new features	×	×	×
	Granular permissions for Dynamic Data Masking	×	×	×
	Data Lake Virtualization	×	×	×
Analytics	Object Storage Integration with S3-compatible storage		×	×

Other specialty editions of SQL Server 2022 include Developer Edition, which is licensed for non-production use; the freely downloadable and distributable Express Edition. SQL Server Web Edition remains available only under the Microsoft Services Provider License Agreement (SPLA).

- For general information on SQL Server editions reference <https://aka.ms/SQLServer2022EditionsDatasheet>
- For detailed product specifications and a full feature-by-feature comparison of the SQL Server editions, visit <https://aka.ms/sqlserver2022editions>

SQL Server 2022 Edition	Database Engine (DBE) capacity limits			Analysis Services (AS) and Reporting Services (RS) capacity limits*		
	Max compute capacity	Max memory utilization - DBE	Max DB Size	Max compute capacity	Max memory utilization - AS	Max memory utilization - RS
Enterprise Per Core	OS max	OS max	524 PB	OS max	OS max	OS max
Standard Per Core	Lesser of 4 sockets or 24 cores	128 GB	524 PB	Lesser of 4 sockets or 24 cores	64 GB (MOLAP) 16 GB (Tabular)	64 GB
Standard Server + CAL	24 core limit	128 GB	524 PB	Lesser of 4 sockets or 24 cores	64 GB (MOLAP) 16 GB (Tabular)	64 GB
Express	Lesser of 1 socket or 4 cores	1410 MB	10 GB	Lesser of 1 socket or 4 cores	N/A	4 GB (Advanced Services Ed.)
Developer	OS max	OS max	524 PB	OS max	OS max	OS max

This table shows a comparison of the key capacity limits across the SQL Server editions.

*Analysis Services and Reporting Services are available for Windows Server only

- For more information on the compute capacity limits for each edition of SQL Server, visit <https://aka.ms/sqlcapacitylimits>.

SQL Server and Software Assurance

Software Assurance (SA) for Volume Licensing helps boost IT productivity by enabling customers to get the most from Microsoft software products. SA benefits—including 24x7 support, deployment planning services, user and technical training, and the latest software releases and unique technologies—are combined in one cost-effective program.

For SQL Server, using these benefits can help customers improve productivity and help IT efficiently deploy and manage SQL Server software. As hardware capacity and licensing needs expand, SQL Server customers with SA coverage can enjoy the benefit of adding incremental licenses without worrying about the software version licensed. Software licenses and use rights are version-specific and as such, licenses for different software versions cannot be combined when licensing a single operating system environment. As a benefit of having access to the latest version of SQL Server software, SA customers licensed under the core licensing model (for example) can easily combine current version core licenses with future version core licenses, without the need to track or otherwise reassign covered licenses based on software version alone.

Customers purchasing subscription licenses through the Microsoft Customer Agreement (MCA) benefit from some of the same rights as customers with Software Assurance, such as access to the latest versions, fail-over rights, unlimited virtualization, the Flexible Virtualization Benefit, the option to license by virtual machine and the Azure Hybrid Benefit, see the Product Terms for details. Subscription licenses through the MCA do not include License Mobility through SA.

Note: All licenses must be covered with SA and product use rights do not change when using downgrade rights to deploy prior software versions.

Software Assurance benefits overview	
Benefit	Description
Azure Hybrid Benefit	Azure Hybrid Benefit for SQL Server is an Azure-based benefit that enables customers to use SQL Server licenses with Software Assurance or subscription licenses to pay a reduced rate ("base rate") on Azure SQL Database vCore-based options such as managed instance, vCore-based single database, and vCore-based elastic pool; on SQL Server in Azure Virtual Machines (including, but not limited to, Azure Dedicated Host); and on SQL Server Integration Services.
Fail-Over servers for disaster recovery ^{New}	Allows customers to install and run passive SQL Server instances in a separate OSE or server for disaster recovery in anticipation of a failover event
Fail-Over servers for disaster recovery in Azure ^{New}	Allows customers to install and run passive SQL Server instances in a separate OSE or server for disaster recovery in Azure in anticipation of a failover event
Fail-Over servers for high availability	Allows customers to install and run passive SQL Server instances in a separate OSE or server for high availability in anticipation of a failover event
Unlimited virtualization (VMs)	Allows customers to run any number of instances of SQL Server Enterprise Edition software in an unlimited number of VMs. Applicable under the core licensing model only.
Unlimited containers	Allows customers to run any number of containers in a virtual OSE when licensing by virtual OSE (virtual core licensing). Available for Standard or Enterprise licenses.
Flexible Virtualization Benefit	Allows assignment of subscription licenses or licenses with Software Assurance to Authorized Outsourcers' servers (shared or dedicated) under expanded Outsourcing Software Management rights.
Power BI Report Server	Allows SQL Server Enterprise Edition customers to run Power BI Report Server.
Option to license by virtual machine	With subscription licenses or licenses with Software Assurance, SQL Server 2022 may be licensed by virtual machine, and licenses may move with the virtual machine at any time as needed to another server in the same server farm.

License Mobility within a server farm	Allows reassignment of SQL Server licenses within a server farm more than once every 90 days. Does not apply to SQL Server PDW.
License Mobility through SA	Allows license reassignment of SQL Server to Authorized License Mobility Partners' shared servers. Does not apply to SQL Server PDW.
Additional benefits for SCE customers	In addition to the benefits noted above, Server Cloud Enrollment (SCE) customers may also qualify for premium benefits, including Unlimited Problem Resolution Support.

This table provides an overview of the benefits of Software Assurance.

For example, by combining SQL Enterprise Edition with the benefits provided through SA (or subscription licenses), customers unlock the full power of SQL Server:

- Stay current with all SQL Server features
- Access an unlimited number of virtual machines
- Modernize to the cloud with existing licenses
 - Take advantage of high availability and Disaster recovery scenarios at no additional licensing cost
- Generate data visualizations on premises with Power BI Report Server

Refer to the Volume Licensing Product Terms for more details on these benefits and additional license grants available to SQL Server customers with SA, including any additional terms and conditions that may apply.

- For more information on Software Assurance benefits, visit <https://aka.ms/SQLServerSoftwareAssurance>

How SQL Server licenses are sold

SQL Server software licenses are sold through channels designed to meet the unique needs of customers. These sales channels include online retailers offering full packaged product (FPP) licenses of SQL Server software, Original Equipment Manufacturers (OEMs) offering pre-installed licenses with their hardware systems, as well as Licensing Solutions Partners (LSPs) and Enterprise Software Advisors (ESAs) offering SQL Server software through Microsoft Volume Licensing programs for end-customer organizations.

For customers with as few as five users, Microsoft offers licensing programs to help reduce administrative overhead and software management costs, while enabling product licensing on an ongoing basis at a considerable discount. The various licensing options enable customers to choose the program that works best for their management and operational needs.

- Comprehensive programs that offer Software Assurance as a fixed benefit include the Open Value (OV), Open Value Subscription (OVS), Enterprise Agreement (EA), Enterprise Subscription Agreement (EAS) and the Server and Cloud Enrollment (SCE).
- Transactional programs include Open* and the Microsoft Products and Services Agreement (MPSA).
- SQL Server Subscriptions are also offered through Cloud Server Provider (CSP channel)
- Beginning with SQL Server 2022, customers may purchase SQL Server on-premises or in third party clouds through Microsoft Azure with Azure Arc-enabled SQL Server on a consumption base.

Server and Cloud Enrollment

The Server and Cloud Enrollment (SCE) is an enrollment under the Microsoft Enterprise Agreement that enables highly committed customers to standardize broadly on one or more key server and cloud technologies from Microsoft. In exchange for making an installed base-wide commitment to one or more components of the Server and Cloud Enrollment, customers receive the best pricing and terms, plus other benefits, including cloud-optimized licensing options and simplified license management.

Microsoft also offers programs that can meet the specific needs of organizations that partner with Microsoft to provide additional software and services, such as the Microsoft Independent Software Vendor (ISV) Royalty Licensing Program and the Microsoft Services Provider License Agreement (SPLA).

SQL Server	Retail	Volume licensing programs				Third party	
	FPP/ESD	OPEN*	MPSA	CSP	EA/EAS/SCE	ISVR	SPLA
Enterprise Edition		•	•	•	•	•	•
Standard Edition	•	•	•	•	•	•	•
Web Edition							•
Express Edition		Free download					
Developer Edition		Free download					

This table shows the primary channel availability for SQL Server software licenses. Every edition may not be available in all channels or licensing programs in all regions. *New licenses are no longer available through the Open License program. For upgrade rights under active Software Assurance, existing customers should refer to their agreement.

- For more information about Microsoft Volume Licensing Programs, download the Volume Licensing Reference Guide at <https://aka.ms/VLReferenceGuide>
- For details on the Microsoft Server and Cloud Enrollment, visit <https://aka.ms/EnterpriseLicensing>

SQL Server licensing models

Microsoft offers a variety of SQL Server licensing options aligned with how customers typically purchase specific workloads. The Server+CAL licensing model provides the option to license users and/or devices and then have low-cost access to incremental SQL Server deployments. For customers who cannot count users or require premium database capabilities, Microsoft licenses SQL Server in a core-based licensing model. Core-based licensing gives customers a more precise measure of computing power and a more consistent licensing metric, regardless of whether solutions are deployed on physical on-premises servers, or in virtual or cloud environments.

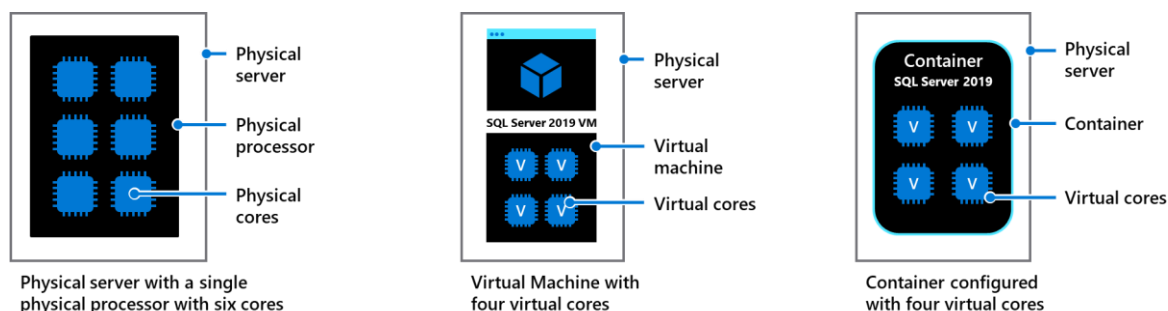
SQL Server	Description	Licensing options		
		Server+CAL	Per Core	Requirements
Enterprise	For applications requiring mission critical in-memory performance, security and high availability		•	
Standard	Delivers fully featured database capabilities for mid-tier applications and data marts	•	•	SQL Server CALs required when licensing Server+CAL

This table compares the licensing options for each of the main SQL Server editions.

Core-based licensing

Under the Per Core licensing model, **each server** running SQL Server software or any of its components (such as Reporting Services or Integration Services) must be assigned an appropriate number of SQL Server core licenses. The number of core licenses needed depends on whether customers are licensing the physical server based on physical cores or licensing individual virtual operating system environments (OSEs).

Unlike the Server+CAL licensing model, the Per Core model allows access for an unlimited number of users or devices to connect from either inside or outside an organization's firewall. With the Per Core model, customers do not need to purchase additional client access licenses (CALs) to access the SQL Server software.



This figure depicts the representations of the physical server, physical processor, physical and virtual cores, virtual machines and containers that are used in this guide.

Physical server	A server is a physical hardware system capable of running server software. A hardware partition or blade is considered to be a separate physical hardware system.
Physical processor	A processor is generally a physical chip that resides in a physical socket of the hardware partition and contains one or more cores.
Physical core	Each physical processor contains smaller processing units called physical cores . Some processors have two cores, some four, some six or eight, and so on.
Virtual core	A virtual core is a virtual representation of one or more hardware threads.
Hardware thread	A hardware thread is either a physical core or a hyper-thread in a physical processor.
Physical Operating System Environment	A physical operating system environment (OSE) is configured to run directly on a physical hardware system and is all or part of an operating system instance.

For detailed definitions of these and other key licensing terms, please refer to the Microsoft Volume Licensing Product Terms.

How to license SQL Server using the per core licensing model

When running SQL Server in a **physical OSE**, all physical cores on the server must be licensed. Software partitioning does not reduce the number of core licenses required, except when licensing individual virtual machines (VMs). **A minimum of four core licenses is required for each physical processor on the server.**

To determine and acquire the correct number of core licenses needed, customers must:

1. Count the total number of physical cores in the server.
2. Purchase the appropriate number of core licenses required for the server. Core licenses are sold in packs of two, so customers must divide the number of licenses required by two to determine the actual number of line items (licensing SKUs) to order.

For more details on the per core licensing model, including key terms and licensing definitions, download the Introduction to per core Licensing Volume Licensing Brief <https://aka.ms/LicensingByCores>.

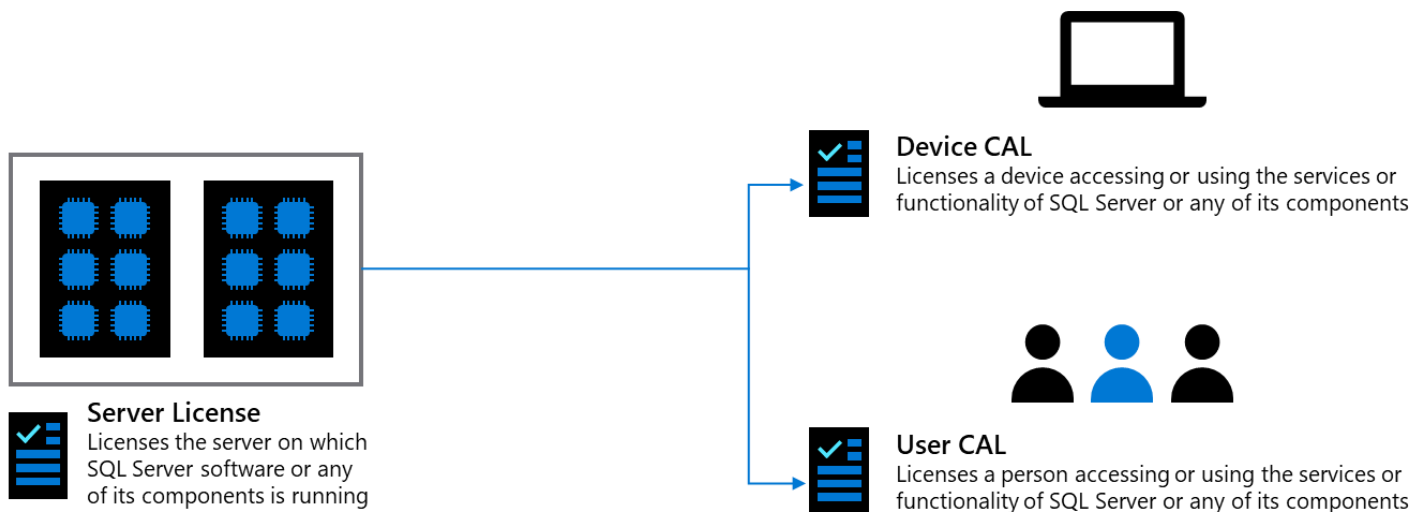
The per core licensing model is appropriate when:

- Deploying the SQL Server Enterprise Edition (including using the SQL Server Parallel Data Warehouse deployment option) or SQL Server Web Edition software.
- Deploying internet or extranet workloads, systems that integrate with external-facing workloads (even if external data goes through one or more other systems), or when the number of users/devices cannot be counted easily.
- Implementing centralized deployments that span across a large number of direct and/or indirect users/devices.
- The total licensing costs for licensing SQL Server Standard Edition software are lower than those incurred using the Server+CAL licensing model.

Note: The use of hyper-threading technology does not affect the number of core licenses required when running SQL Server software in a physical OSE.

Server+CAL licensing

When licensing the SQL Server Standard Edition software under the Server+CAL model, customers purchase a server license for each server, and a client access license (CAL) for each device (Device CAL) and/or user (User CAL) accessing SQL Server or any of its components. A CAL is not software; it is a license granting users and devices access to the SQL Server software.



This figure illustrates the licenses used in the Server+CAL licensing model.

How to License SQL Server Using the Server+CAL Licensing Model

Under the Server+CAL licensing model, each operating system environment (OSE) running SQL Server software or any of its components must have a SQL Server server license assigned to the physical server hosting the OSE. Each server license allows customers to run any number of SQL Server instances in a single OSE, either physical or virtual.

Running SQL Server software on different hardware partitions or blades requires separate software licenses. Hardware partitions and blades are considered to be separate servers for licensing purposes and SQL Server software licenses cannot be assigned to more than one server at any time.

Devices not operated by humans require device CALs, even when connecting to SQL Server indirectly. For human operated devices such as PCs or hand-held terminals, a user CAL or device CAL can be used. CALs must be for the version of SQL Server being accessed or a later version.

Note: The use of hardware or software that reduces the number of devices or users that directly access or use the software (multiplexing/pooling) does not reduce the number of CALs required. For details on how to license SQL Server in a multiplexed application environment, refer to the [Advanced licensing scenarios](#) section of this guide.

Licensing SQL Server components

SQL Server software includes a range of licensed server components, including but not limited to the SQL Server Database Engine (DB), Master Data Services (MDS), Analysis Services (AS), Integration Services (IS), Reporting Services (RS), and Data Quality Services (DQS). In addition, a number of management components, such as client applications and tools used for creating or working with analytical data, are provided.

For more details on the software components specifically included with SQL Server, visit <https://aka.ms/sqlserver2022editions>. The software components of a single SQL Server license cannot be separated. Any OSE running any of the licensed components of SQL Server, requires a license. For example, if the SQL Server DB is deployed in one OSE and SQL Server RS is deployed in another, both OSEs must be fully licensed for SQL Server accordingly.

Management tools and other software identified as additional or supplemental software—such as product documentation, client connectivity tools, software add-ins, and Software Development Kits (SDKs)—can generally be distributed and run on any number of devices for use with a licensed instance of SQL Server software. Refer to the Volume Licensing Product Terms for the list of additional software components provided with SQL Server.

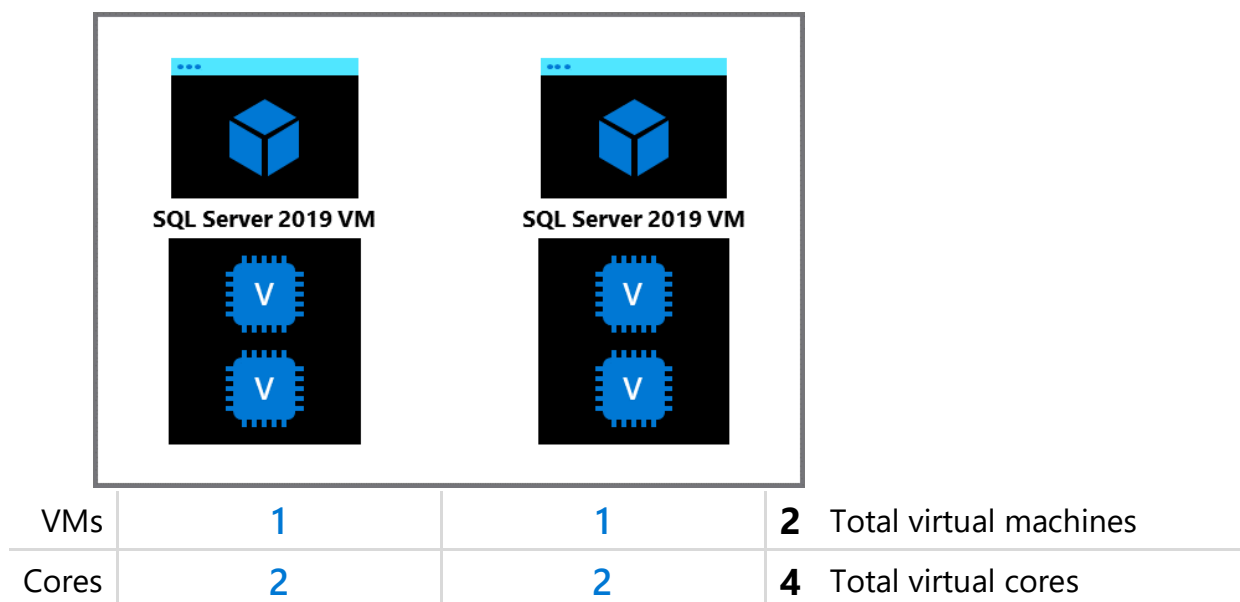
Licensing SQL Server in a virtualized environment

Microsoft SQL Server is increasingly being deployed in virtualized environments, which enable running instances of SQL Server concurrently in separate virtual OSEs (or virtual machines).

SQL Server offers virtualization rights, options and benefits to provide flexibility for customers deploying in virtual environments. When deploying SQL Server software in virtualized environments, customers have the choice to license either individual virtual machines as needed, or to license for maximum virtualization in highly virtualized, private cloud, or dynamic environments.

Note: Beginning with SQL Server 2022, licensing by virtual machine is an option under subscription licenses or licenses with Software Assurance only.

When licenses are allocated by virtual machine, they can be moved at any time as needed to another server within the same Server Farm (e.g., with the virtual machine). (The right to reassign licenses to servers in a different Server Farm is subject to the standard 90-day reassignment limitation.) Maximum virtualization can be achieved by licensing the entire physical server with Enterprise Edition core subscription licenses or licenses with Software Assurance (SA).



This figure depicts two virtual machines, each containing two virtual cores.

Licensing individual virtual machines

As customers consolidate existing workloads and refresh hardware, they may find that a SQL Server instance uses only a fraction of available system computing power. When deploying databases in virtual environments that require just a fraction of a physical server, savings can be achieved by licensing individual virtual machines (VMs).

How to license individual virtual machines using the Per Core licensing model

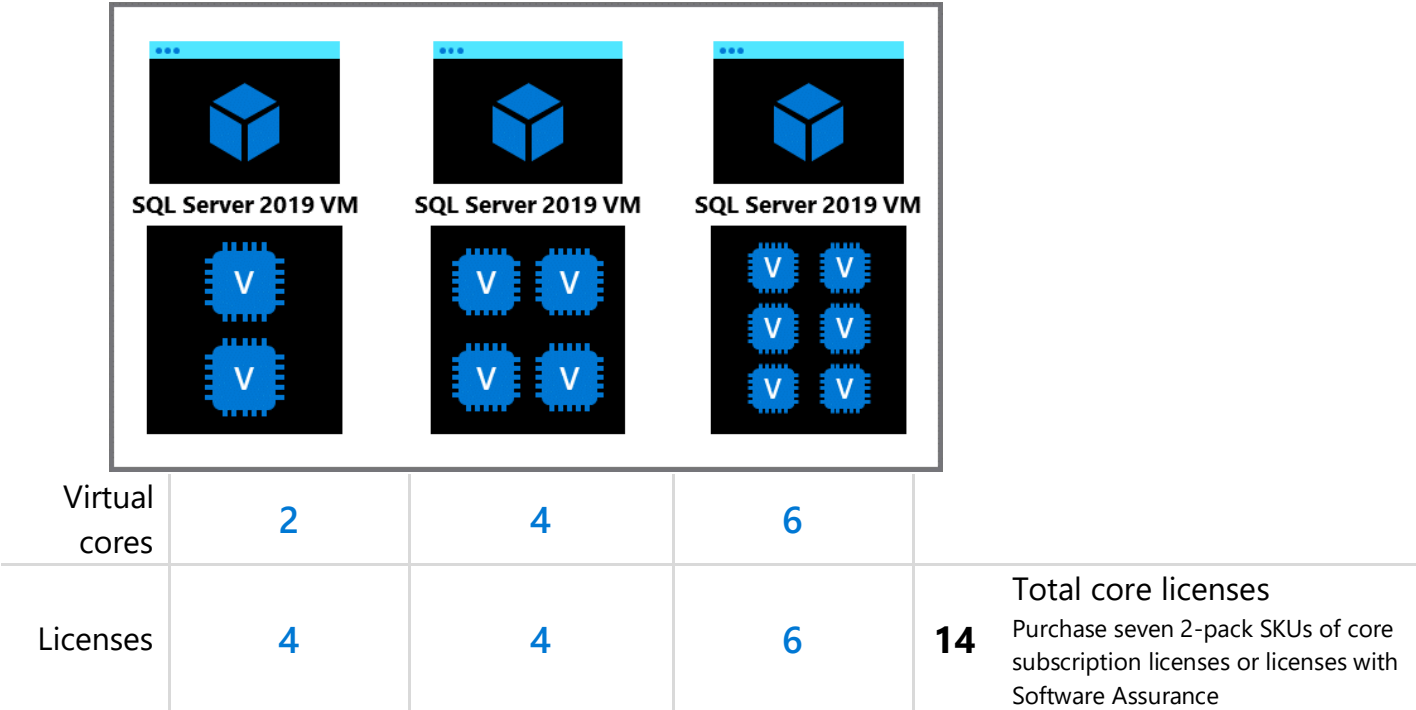
Similar to the Per Core licensing model in physical OSEs, all virtual cores (v-cores) supporting virtual OSEs that are running instances of SQL Server software must be licensed accordingly.

To license individual VMs using the Per Core model, customers must purchase a core subscription license or license with Software Assurance for each v-core (or virtual processor, virtual CPU, virtual thread) allocated to the VM, subject to a four-core license minimum per VM. For licensing purposes, a v-core maps to a hardware thread.

Note: Licensing individual VMs is the only licensing option available for SQL Server Standard Edition customers who are running the software in a virtualized environment under the Per Core model.

When licensing by virtual machine (as described above) customers may move licenses at any time as needed to another server within the same Server Farm. For more information on licensing for application mobility, refer to the [Advanced licensing scenarios](#) section of this guide.

- 1. License the virtual cores in each virtual machine
- 2. There is a minimum of four core subscription licenses or licenses with Software Assurance required for each virtual machine



This figure illustrates the licensing requirements for three different virtual machines under the Per Core licensing model.

Additional licenses are required when:

- A single hardware thread supports multiple virtual cores. (A core subscription license or license with Software Assurance is required for each v-core.)
- Multiple hardware threads support a single virtual core simultaneously. (A core license allows a single v-core to be supported by a single hardware thread.)

How to license individual virtual machines using the Server+CAL licensing model

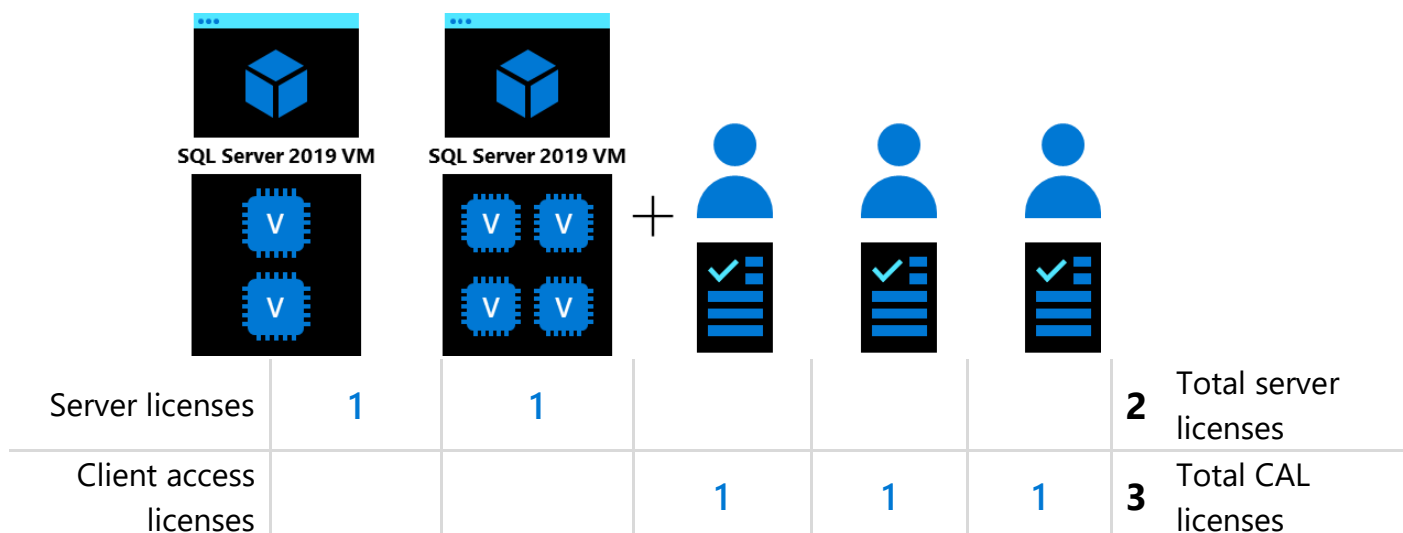
To license individual VMs using the Server+CAL model, customers simply purchase one server license for each VM running SQL Server software, regardless of the number of virtual processors allocated to the VM.

For example, a customer who wants to deploy Standard Edition running in six VMs, each allocated with four v-cores, would need to assign six SQL Server Standard server licenses to that server.

Note: Each user or device accessing SQL Server software, regardless of a virtual or physical deployment, requires a SQL Server CAL (for the same or a later version).

For details on how to license individual VMs with legacy SQL Server Enterprise Edition server licenses, please refer to the [Additional product information](#) section of this guide.

1. License each virtual machine with a server license
2. License each user or device with a CAL (for the same or a later version)



This figure shows an example of licensing virtual machines under the Server+CAL licensing model.

The right to use SQL Server in a virtual machine under the Server+CAL licensing model does not require subscription licenses or licenses with Software Assurance; however, with subscription licenses or licenses with Software Assurance, a customer using SQL Server in a virtual machine may move licenses at any time as needed (e.g., with the virtual server) to another server in the same Server Farm.

Licensing for maximum virtualization

With SQL Server Enterprise Edition, customers who have **licensed all physical cores on the server** can run an unlimited number of instances of the software in a number of OSEs/containers (physical and/or virtual) equal to the number of core licenses assigned to the server. For example, a four-processor server with four cores per processor—fully licensed with 16 core licenses—can run SQL Server software in up to 16 VMs, regardless of the number of virtual cores allocated to each VM.

- Customers who have licensed all the physical cores on the server and want to run SQL Server software in more VMs than are permitted, can assign additional core licenses to the licensed server.
- Each additional core license allows deployment of SQL Server software in an additional VM, so in the previous example, a customer who wants to run SQL Server Enterprise Edition in 18 VMs would simply acquire and assign 18 core licenses to that server.

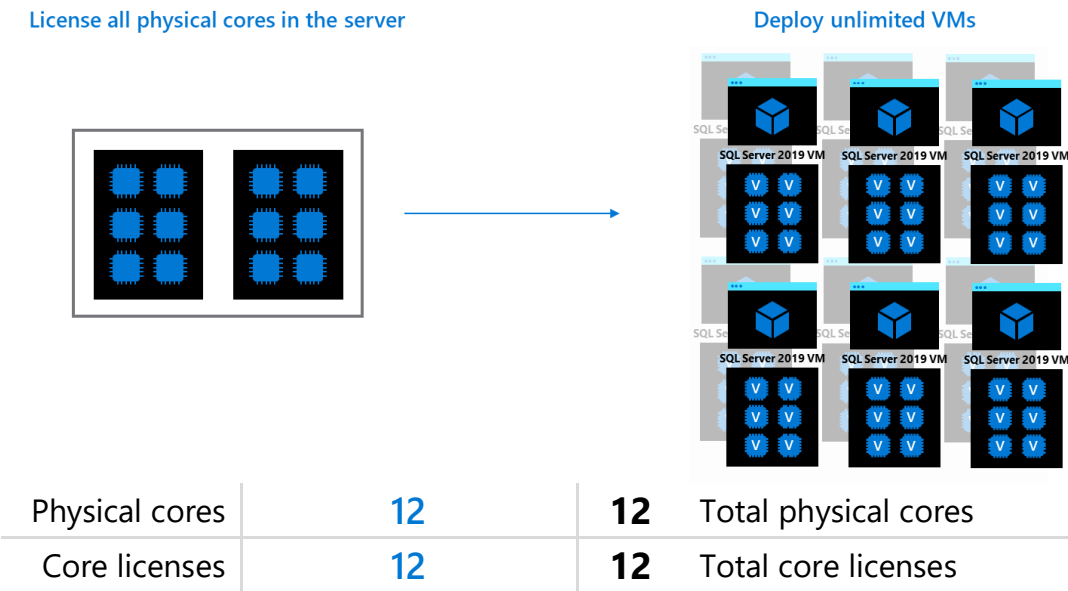
On servers fully licensed (i.e., based on physical cores) under SQL Server Enterprise subscription licenses or licenses with Software Assurance (SA), customers' use rights allow any number of instances of the software to

run in any number of OSEs/containers (physical or virtual). This benefit enables customers to deploy an unlimited number of VMs to handle dynamic workloads and fully utilize hardware computing capacity.

Note: This benefit ends when the subscription licenses or SA coverage expire(s).

Licensing for maximum virtualization can be an ideal solution when:

- Deploying SQL Server private cloud scenarios with high VM density
 - Hyper-threading is being used
 - Using dynamic provisioning and de-provisioning of VM resources
1. Fully license the physical server with SQL Server Enterprise Edition core subscription licenses or licenses with Software Assurance
 2. Deploy an unlimited number of virtual machines



Shown is an example of licensing a 12-core physical server for unlimited VMs with Enterprise Edition core subscription licenses or licenses with SA.

Licensing SQL Server in containers

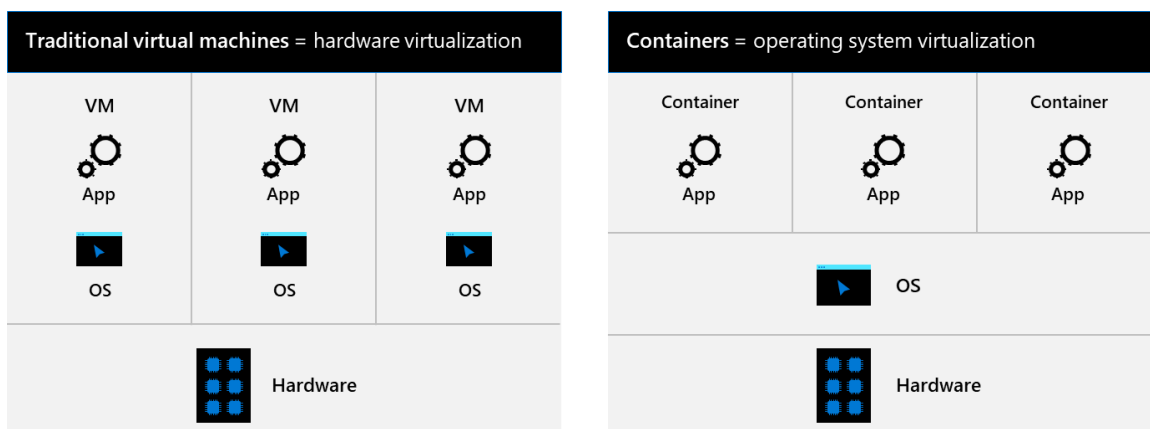
SQL Server can be deployed on Linux and Docker platforms. This allows customers to not only choose their platform, but also to deploy SQL Server in containers using Docker container technology.

Containers provide operating system level virtualization that allows multiple isolated applications to be run on a single system. Containers make it easy to enable:

- Rapid deployment scenarios
- Separation of application services through microservices deployments
- Rapid scale up and scale down of application services

Containers differ from virtual machines in that they provide operating system-level virtualization and all containers running on a physical host share the operating system. This provides application isolation, but not operating system-level isolation. However, containers are less resource intensive, require much less overhead to run, can be started and scaled rapidly, and provide a high degree of portability.

Virtual machines provide hardware-level virtualization, with each virtual machine having its own operating system environment, virtual memory and virtual cores. This provides full isolation from other virtual machines and allows for unique settings for the operating system within each virtual machine.



This figure depicts the structural differences between containers and virtual machines.

For licensing, an operating system environment is defined as all or part of an operating system instance, or all or part of a virtual operating system instance which enables separate machine identity. Containers and virtual machines are structured differently, but they are considered the same from a licensing perspective.

Licensing individual containers

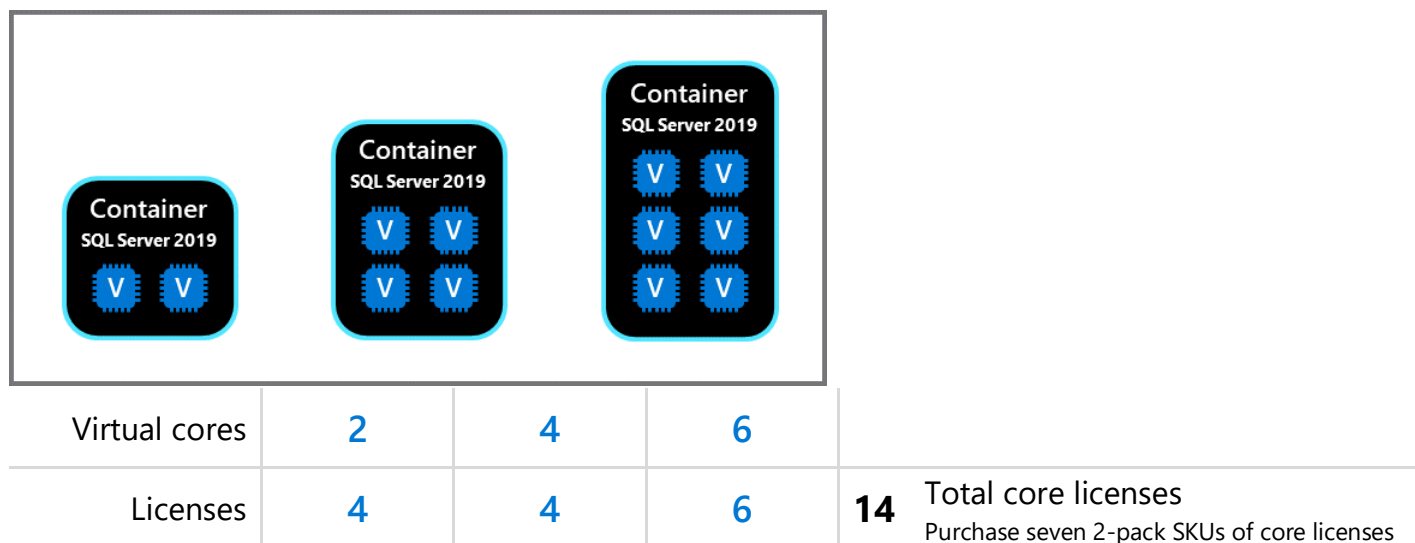
How to license individual containers using the Per Core licensing model

Similar to the Per Core licensing model in physical OSEs, all virtual cores (v-cores) supporting containers running instances of SQL Server software must be licensed accordingly.

To license individual containers using the Per Core model, customers must purchase a core license for each v-core (or virtual processor, virtual CPU, virtual thread) allocated to the container, subject to a four-core license minimum per container. For licensing purposes, a v-core maps to a hardware thread.

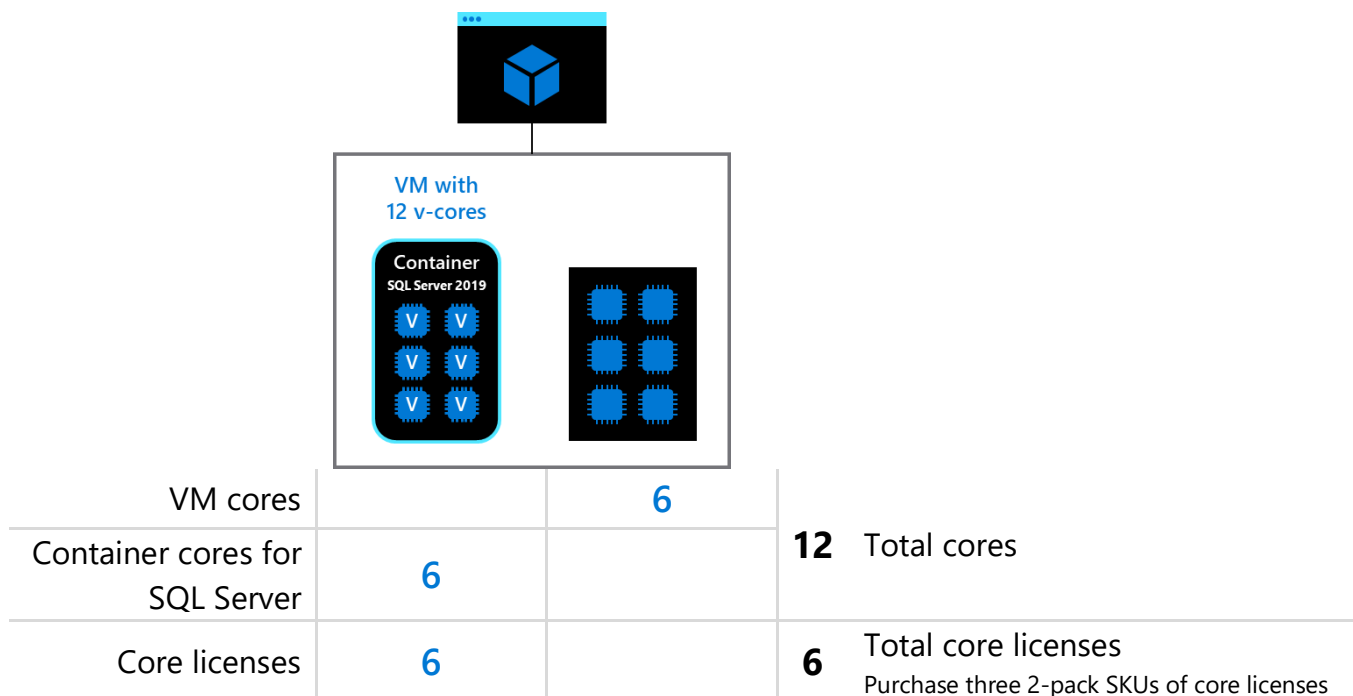
For customers with high density container environments who want to move containers dynamically across servers to reallocate resources as needed, Microsoft offers flexible reassignment rules with the Flexible Virtualization Benefit.

1. License the virtual cores in each container
2. There is a minimum of four core licenses required for each container



This figure illustrates the licensing requirements for three different container configurations under the Per Core licensing model.

To license individual containers deployed using nested virtualization i.e. within a VM, customers must purchase a core license for each v-core (or virtual processor, virtual CPU, virtual thread) allocated to the container, subject to a four-core license minimum per container.



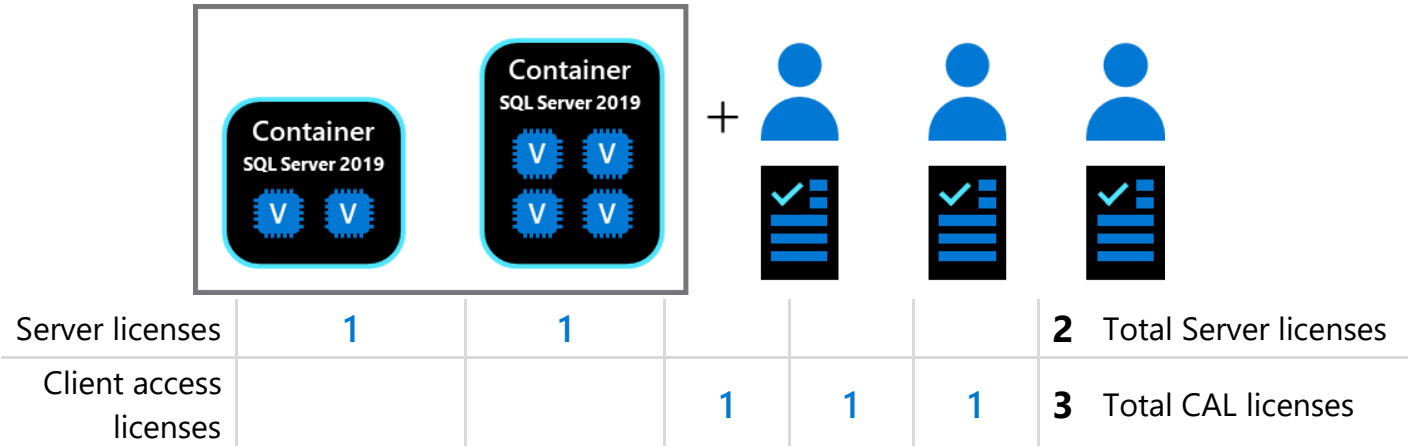
How to license individual containers using the Server+CAL licensing model

To license individual containers using the Server+CAL model customers simply purchase one server license for each container running SQL Server software, regardless of the number of virtual processors allocated to the container.

For example, a customer who wants to deploy Standard Edition running in six containers, each allocated with four v-cores, would need to assign six SQL Server Standard server licenses to that server.

Note: Each user or device accessing SQL Server software, regardless of a virtual or physical deployment, requires a SQL Server CAL (for the same or a later version).

- 1. License each container with a server license
- 2. License each user or device with a CAL



This figure shows an example of licensing containers under the Server+CAL licensing model.

Licensing containers for maximum density

Licensing by physical core – Enterprise only

With SQL Server Enterprise Edition only, customers who have **licensed all physical cores on the server** can run a number of VMs/containers equal to the number of core licenses assigned to the server. For example, a four-processor server with four cores per processor—fully licensed with 16 core licenses—can run SQL Server software in up to 16 containers, regardless of the number of virtual cores allocated to each container.

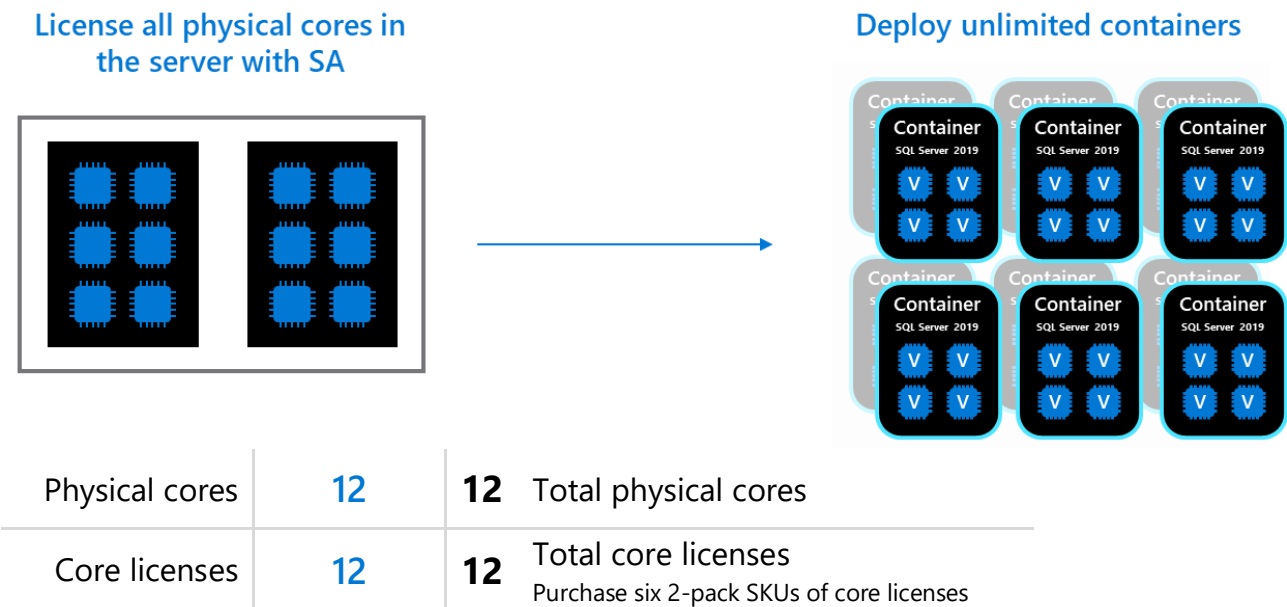
- Customers who have licensed all the physical cores on the server and want to run SQL Server software in more containers than are permitted, can assign additional core licenses to the licensed server.
- Each additional core license allows deployment of SQL Server software in an additional container, so in the previous example, a customer who wants to run SQL Server Enterprise Edition in 18 containers would simply acquire and assign 18 core licenses total to that server.

With the addition of Software Assurance (SA) coverage or a subscription license on all Enterprise Edition core licenses (for a fully licensed server), customers’ use rights are expanded to allow any number of containers to run on the licensed server. This benefit enables customers to deploy an unlimited number of containers to handle dynamic workloads and fully utilize hardware computing capacity.

Note: This benefit ends when SA coverage expires.

Licensing under this model can be an ideal solution when:

- Deploying SQL Server private cloud scenarios with high container density
 - Hyper-threading is being used
 - Using dynamic provisioning and de-provisioning of container resources
1. Fully license the server with SQL Server Enterprise Edition core licenses and Software Assurance
 2. Deploy an unlimited number of containers



Shown is an example of licensing for unlimited containers on 12-core physical server with Enterprise Edition core licenses and SA.

Licensing by virtual core (Virtual OSE) – Standard/Enterprise

With SQL Server Standard or Enterprise Edition **Software Assurance (SA) coverage or subscription licenses**, customers who have **licensed all virtual cores on a virtual OSE** can run any number of containers in the covered virtual OSE. For example, a 16 vCore OSE (including the count of all hyperthreaded cores) can run SQL Server software in any number of containers in that virtual OSE, regardless of the number of virtual cores allocated to each container.

Advanced licensing scenarios and detailed examples

This section introduces a few advanced SQL Server licensing scenarios to help illustrate how customers can apply some of the key licensing principles covered in this guide. For detailed licensing terms and additional licensing guidance applicable to more specific software deployment scenarios, refer to the Microsoft Product Terms.

Licensing SQL Server for high availability and disaster recovery

SQL Server software can be configured so that if one server fails, its processing will be picked up, recovered and continued by another server.

For the purposes of this guide, we are not drawing a distinction in how high availability is being implemented, examples being:

- Log Shipping
- Always On SQL Fail-Over Cluster Instances
- Always On Availability Groups

Each of these implementations uses different terminology. The examples that follow use 'Active' as the read-write database or instance (also referred to as Primary Replica in an Always On Availability Group) and 'Passive' for the write only database or instance (marked to not-read, referred to as Secondary Replica in an Always On Availability Group).

- For additional details on high availability capabilities for SQL Server on Windows visit https://aka.ms/SQLServerRDBMS_Windows
- For additional details on high availability capabilities for SQL Server on Linux visit: https://aka.ms/SQLServerRDBMS_Linux

Failover Basics

For each server OSE licensed with SQL Server subscription licenses or licenses covered by active SA, customers can use the following passive replicas in anticipation of a failover event:

(Note: Failover rights under License Mobility and Azure Hybrid Benefit are different, see below.)

- One passive fail-over replica for High Availability in a separate OSE
- One passive fail-over replica for Disaster Recovery in a separate OSE
- One passive fail-over replica for Disaster Recovery in a single VM or instance on Azure

Note: This benefit applies to all in-market SQL Server versions.

A passive SQL Server replica is one that is not serving SQL Server data to clients or running active SQL Server workloads. The passive failover instances can run on a separate server. These may only be used to synchronize with the primary server and perform the following maintenance-related operations for the permitted passive fail-over Instances:

- Database consistency checks
- Log Back-ups
- Full Back-ups
- Monitoring resource usage data

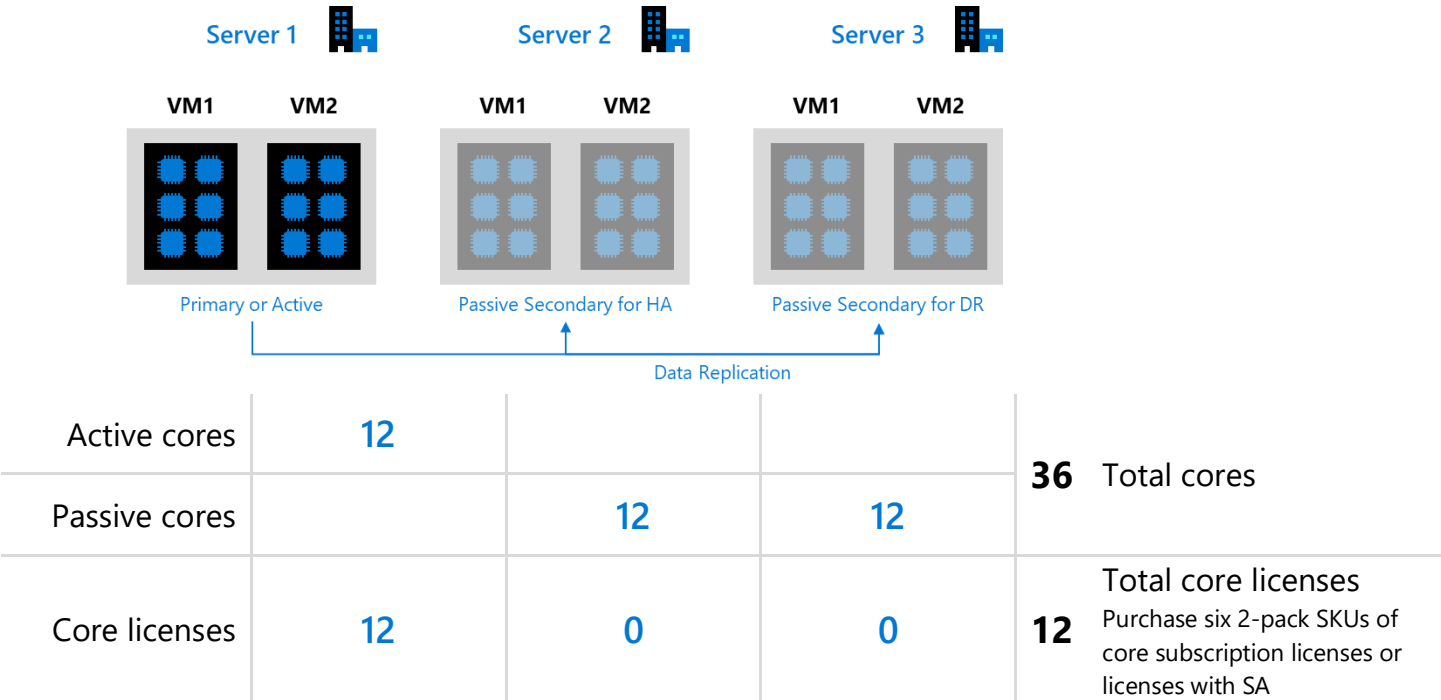
Customer may also run primary and the corresponding disaster recovery replicas simultaneously for brief periods of disaster recovery testing every 90 days.

Note:

- *High Availability replica is defined as a passive replica setup as synchronous replica with automated failover.*
- *Disaster Recovery replica is defined as a passive replica setup as asynchronous replica with manual failover.*

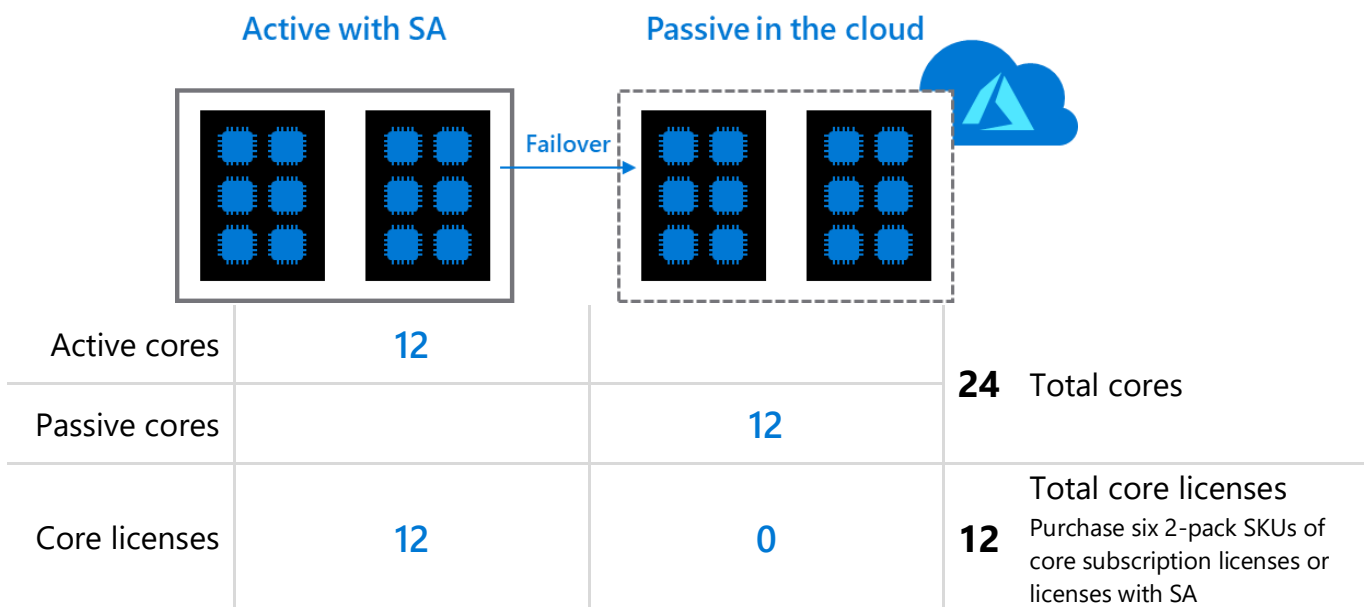
Failover Licensing Scenarios

- Two secondary servers used for failover support do not need to be separately licensed for SQL Server as long as they are passive, and the primary SQL Server is covered with subscription licenses or licenses with active SA. If a server is marked to read and serving data, such as reports to clients running active SQL Server workloads, or performing any “work” except for maintenance related operations mentioned above, then it must be licensed for SQL Server.



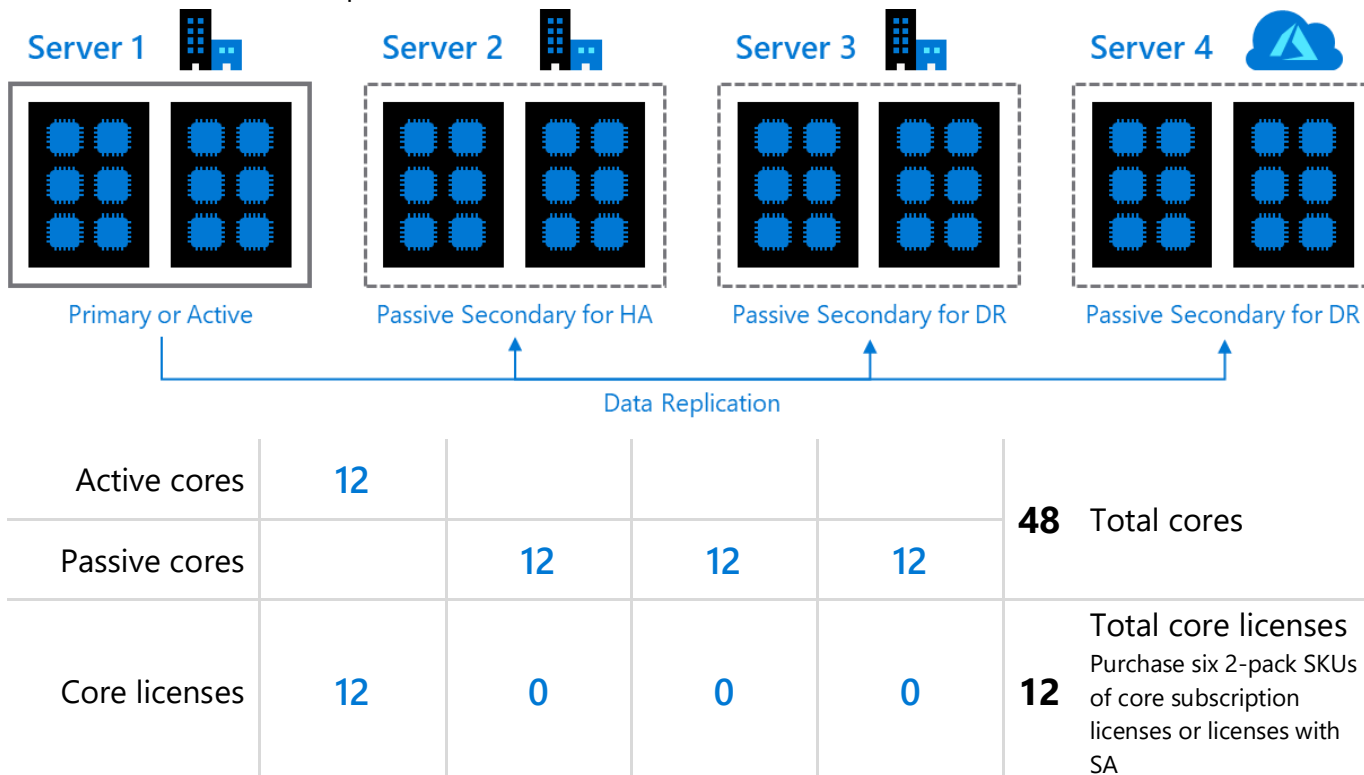
This figure shows an example of licensing an active primary SQL Server database and passive secondary databases hosted on-premises.

- The secondary passive server can also be setup in Azure for Disaster recovery with no additional SQL Server licenses required. Customers will only have to cover the compute costs for VM/server used for passive replica. Additional guidelines to setup DR in Azure can be found at https://aka.ms/Site_Recovery_SQLServer.

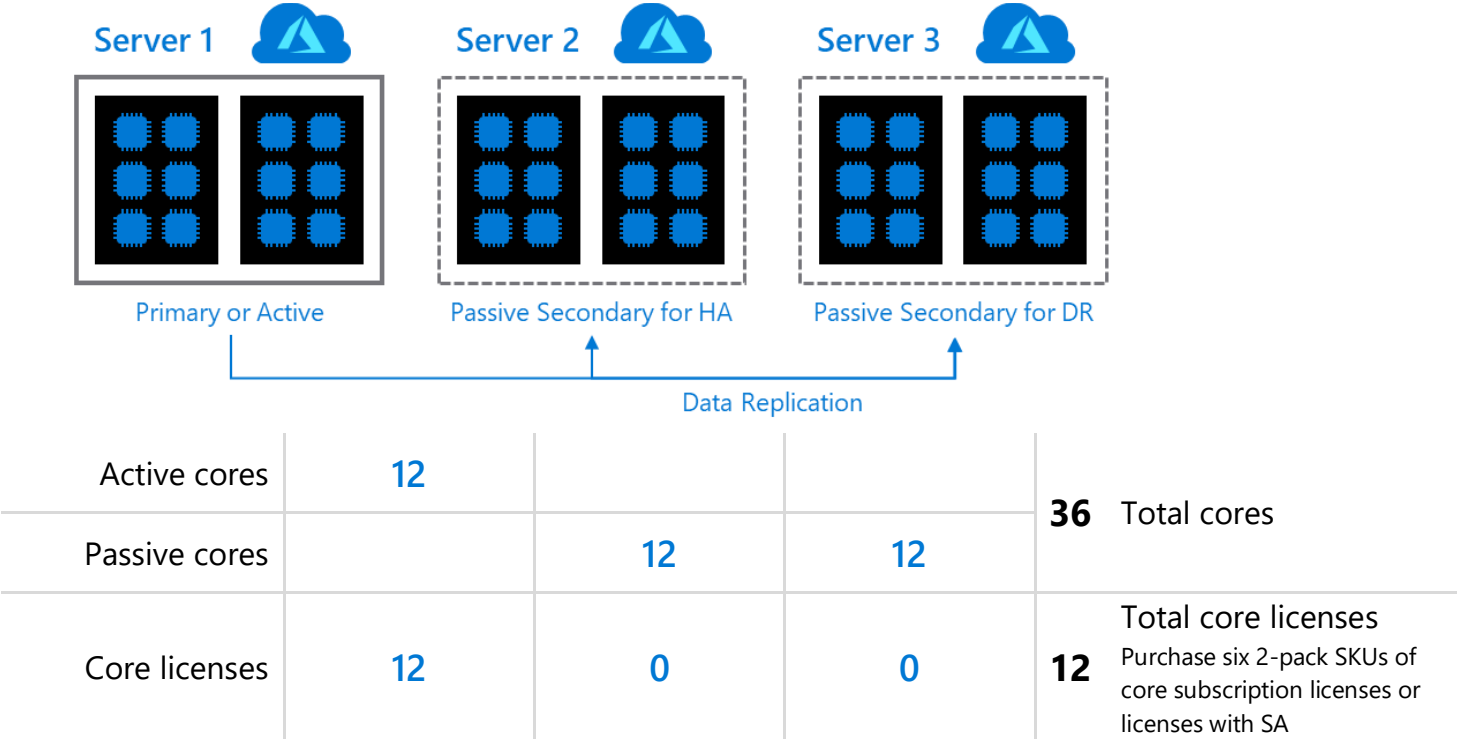


This figure shows an example of licensing an active primary SQL Server database and a passive secondary database hosted in the cloud.

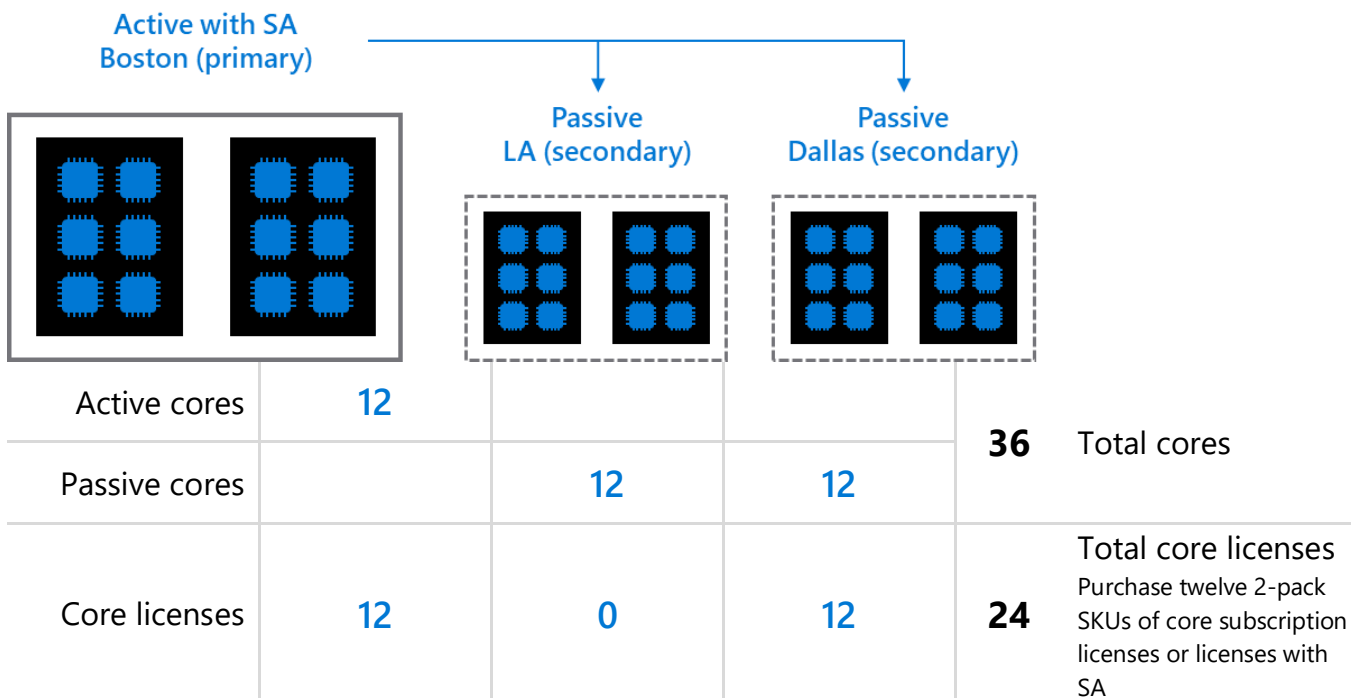
- All the three passive secondary replica benefits can be used simultaneously as long as the on-prem primary is covered with subscription licenses or licenses with SA.



- In the case where a customer is using AHB to license its primary database running on shared hardware in Azure, the customer may run the two passive SQL Server instances (one for HA and one for DR) in a separate OSE running in Azure on shared hardware to support failover events.



- Primary server subscription licenses or licenses covered with SA include support for one Disaster Recovery secondary server only (outside Azure). Any additional secondary Disaster Recovery servers must be separately licensed for SQL Server. Note: The rights to run a passive instance of SQL Server for failover support are not transferable to other licensed servers for purposes of providing multiple passive secondary servers to a single primary server.



This figure shows an example of licensing an active primary SQL Server database and multiple passive secondary databases.

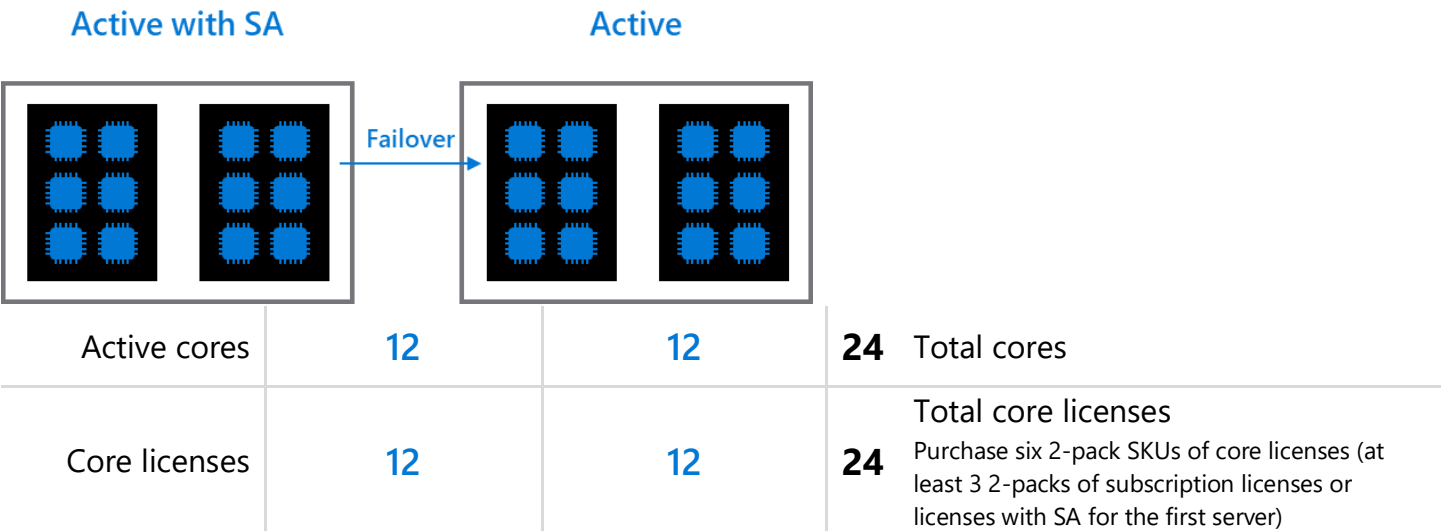
Additional considerations for high availability:

- When licensing SQL Server under the Per Core model, the number of core licenses must be based on the server that requires the higher number of licenses. This way, when the failover server takes over, it will be adequately licensed. For a passive instance of SQL Server to be properly licensed, it cannot require more core licenses than the licensed primary system.
- In the event that a passive instance of SQL Server becomes active for any reason, the primary SQL Server license is dynamically reassigned to the newly active server via either license reassignment rights (under subscription licenses or licenses with Software Assurance when SQL Server is licensed or deployed by virtual machine) or the License Mobility within Server Farms SA Benefit, and now assumes all active workloads.
- In the case where individual SQL Server instances are failing over independently from each other, each SQL Server OSE running an active SQL Server instance requires separate licenses.

Always On Availability Groups

Always On Availability Groups in SQL Server Enterprise Edition enable customers to configure multiple database instances that will failover as a unit, with support for up to eight active secondary replicas and two synchronous secondary replicas. The ability to use secondary replicas for more than just passive failover support can improve the performance of primary, reporting and backup workloads due to better balancing of workloads across instances, helping to provide better return on hardware investment.

Note: When secondary replicas are actively used to support any additional workload scenarios—that is when the replicas used for failover purposes are no longer truly passive—they must be fully licensed accordingly.



This figure shows an example of licensing a 2-node Always On Availability Group with a readable ‘Active’ primary replica SQL Server database and a readable ‘Active’ secondary replica database.

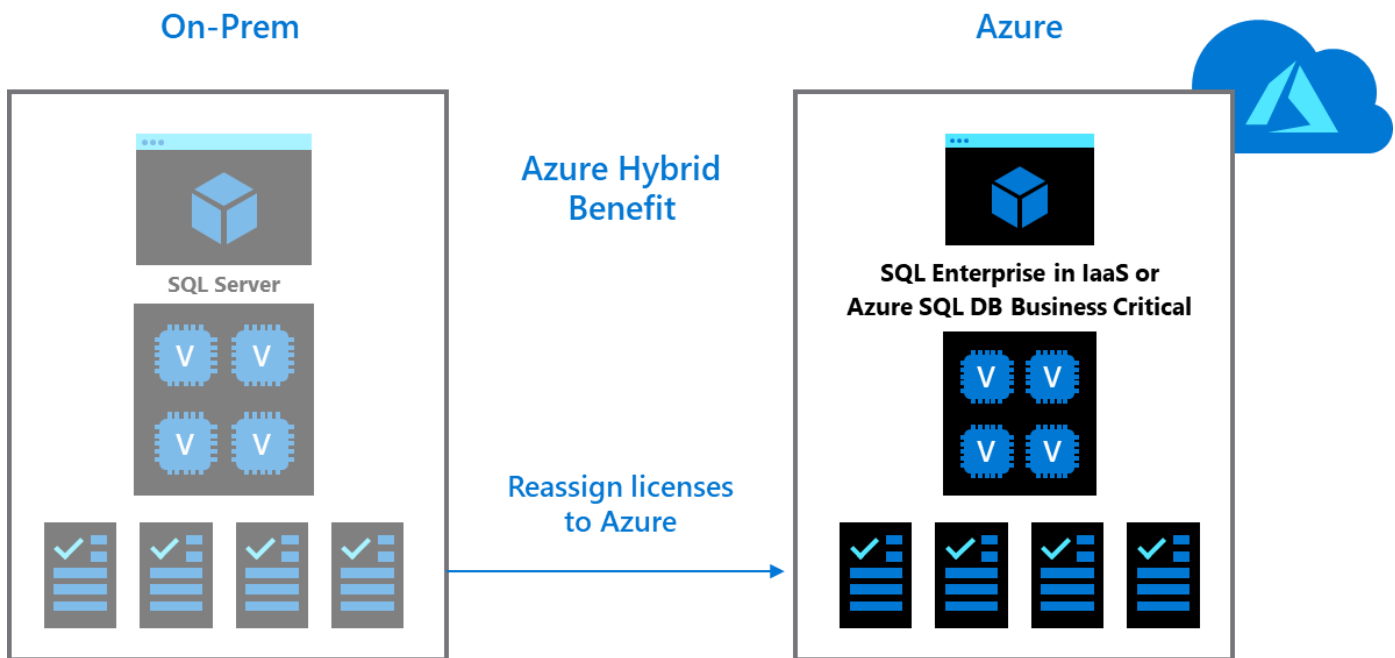
Today, many virtual environments are becoming even more dynamic, especially in scenarios where software is used to automatically and dynamically allocate resources to different VMs “on the fly”. In the next section, we will discuss licensing SQL Server in these scenarios and look at ways to further simplify licensing management.

SQL Server Azure Hybrid Benefit

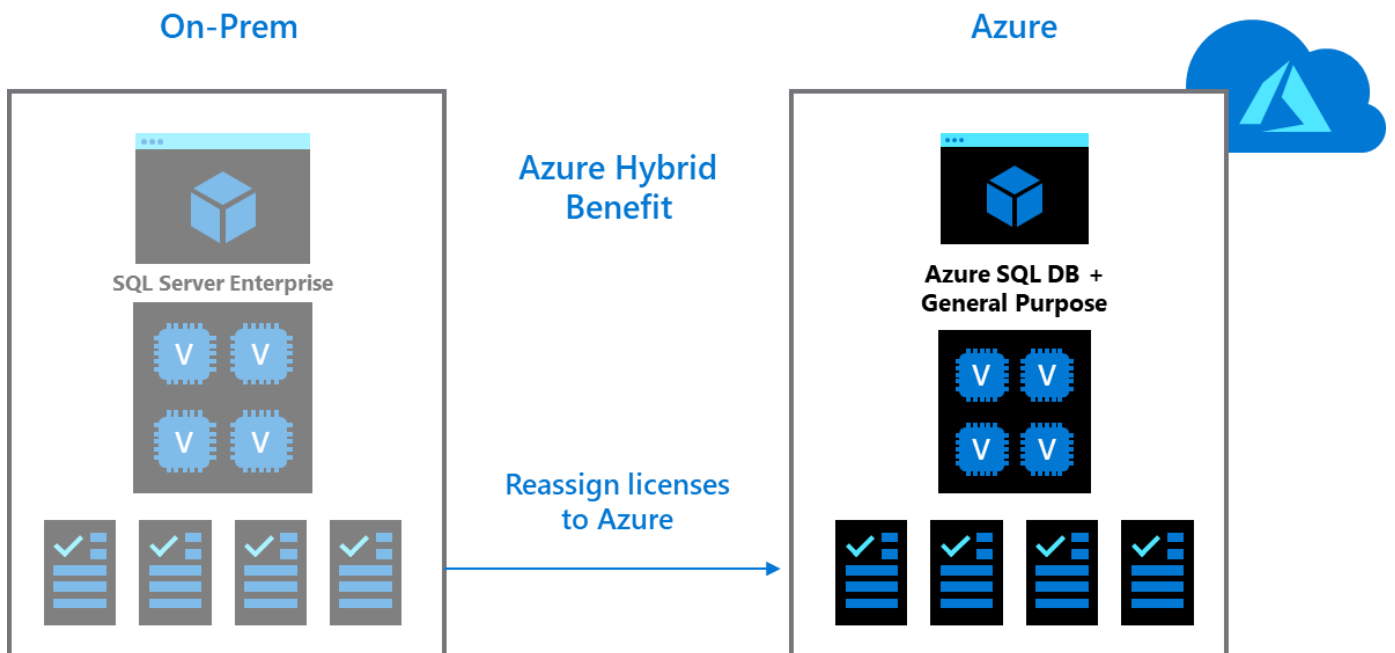
Azure Hybrid Benefit for SQL Server is an Azure-based benefit that enables customers to use SQL Server licenses with Software Assurance or subscription licenses to pay a reduced rate (“base rate”) on SQL Database vCore-based options, such as Managed Instance, vCore-based Single Database, and vCore-based Elastic Pool; on SQL Server in Azure Virtual Machines (including, but not limited to, Azure Dedicated Host); and on SQL Server Integration Services.

For each core of a qualified SQL Server Enterprise core license, customer can use either

- 1 core of Azure SQL Database (Managed Instance, Elastic Pool and Single Database) – Business Critical
- 1 core of Azure Data Factory and SQL Server Integration Services (Standard)
- Or 1 core of SQL Server Enterprise Virtual Machines



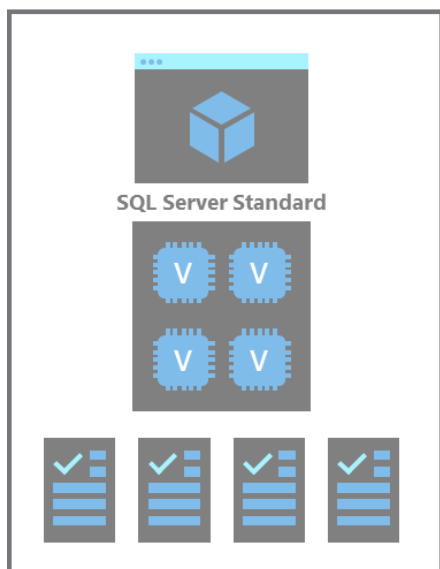
- 4 cores of Azure SQL Database (Managed Instance, Elastic Pool and Single Database) – General Purpose



For each core of a qualified SQL Server Standard core license, customer can use either

- 1 core of Azure SQL Database (Managed Instance, Elastic Pool and Single Database) – General Purpose
- 1 core of Azure Data Factory SQL Server Integration Services (Standard)
- 1 core of SQL Server Standard Virtual Machines

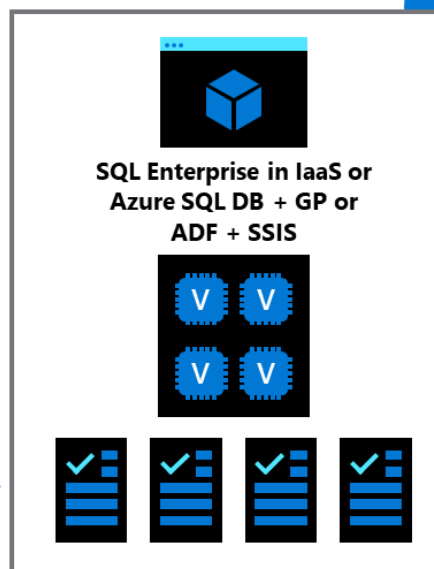
On-Prem



Azure Hybrid Benefit

Reassign licenses
to Azure

Azure



Azure Hybrid Benefit also provides 180 days of dual-use rights to apply to current deployment (on-premises or on third-party cloud) and in Azure to allow for data migration.

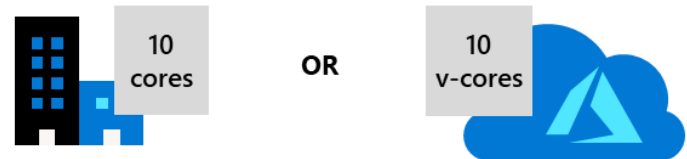
0 days to 180 days

For up to 180 days the 10 cores are used simultaneously on-premises and towards the Azure Hybrid Benefit, allowing time for testing and migrating.



180+ days

After 180 days, the cores can only be used on-premises or in Azure



For a hosted solution with maximum savings, combine the Azure Hybrid Benefit for SQL Server with the Azure Hybrid Benefit for Windows Server.

- For more information on Azure Hybrid Benefit refer to: https://aka.ms/Azure_Hybrid_Benefit

Licensing SQL Server for application mobility

License Mobility is a use right that is available for all editions of SQL Server software licenses with active Software Assurance (SA) coverage. With this SA benefit, customers can reassign SQL Server licenses to different servers within a server farm as often as needed. Customers can also reassign licenses to third party shared servers. License Mobility is available for licenses under both the Per Core and Server+CAL license models.

(Base license terms for subscription licenses and licenses with Software Assurance also permit license reassignment as needed when licensing (or deploying) by virtual machine. This allows customers to move licenses to another server within the same Server Farm at any time as needed. This applies to on-premises deployments and deployments under the Flexible Virtualization Benefit on Authorized Outsourcers' servers (shared or dedicated).)

License Mobility for SQL Server licenses without SA

- SQL Server licenses that are not covered with active SA can only be reassigned to a different server within a server farm once every 90 days, and they cannot be reassigned to a third-party web hoster or non-private cloud at any time. (In the event of permanent hardware failure, the 90-day reassignment limit is waived.) Beginning October 1, 2019, on-premises licenses purchased without Software Assurance and mobility rights can't be deployed with dedicated hosted cloud services offered by "Listed Providers."
- For more information on License Mobility to "Listed Providers", refer to Product Terms. For an FAQ visit <https://aka.ms/outsourcerights>.

License Mobility for SQL Server licenses with SA within or to another server farm

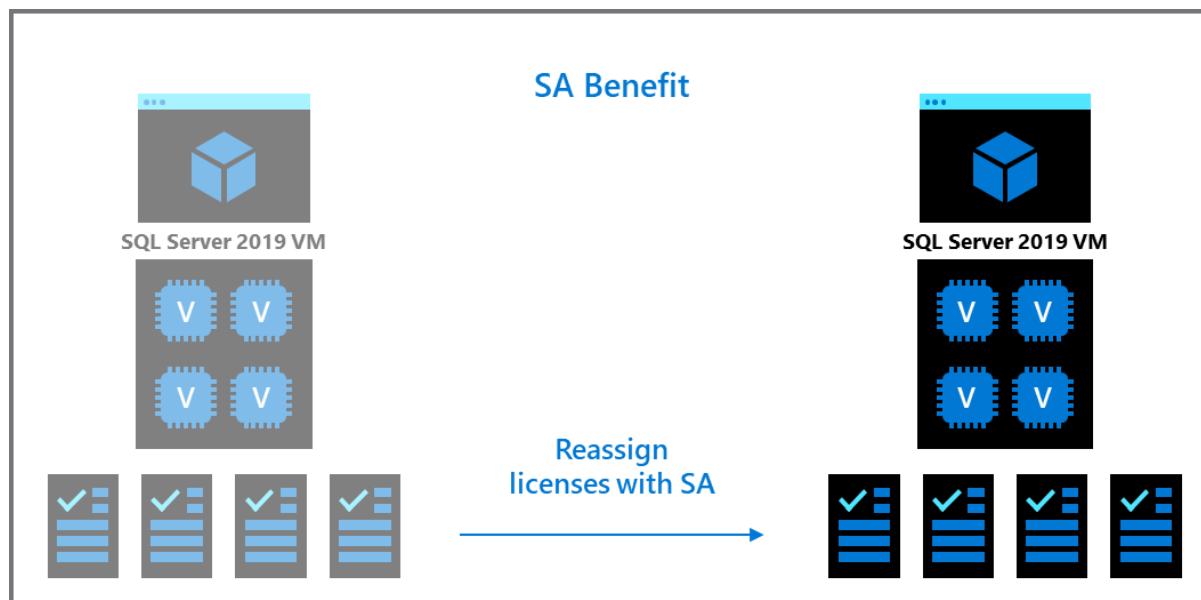
- While SQL Server licenses with active SA can be reassigned to another server within the server farm as often as needed, they can only be reassigned to another server in another server farm, or to a non-private cloud, once every 90 days.

- A server farm may consist of up to two data centers located in time zones that are within four hours of one another and/or with the European Union (EU) and/or European Free Trade Association (EFTA).
- A given data center may only be part of one server farm.

- License Mobility use rights do not apply to SQL Server PDW software.

License mobility rights can benefit customers who license individual virtual machines (VMs) or containers and then want to reassign those licenses to different servers within a server farm—as workloads move dynamically—or to VMs in cloud environments.

Note: License mobility rights apply only to the reassignment of software *licenses* and are not applicable to the reassignment of running instances of SQL Server software.

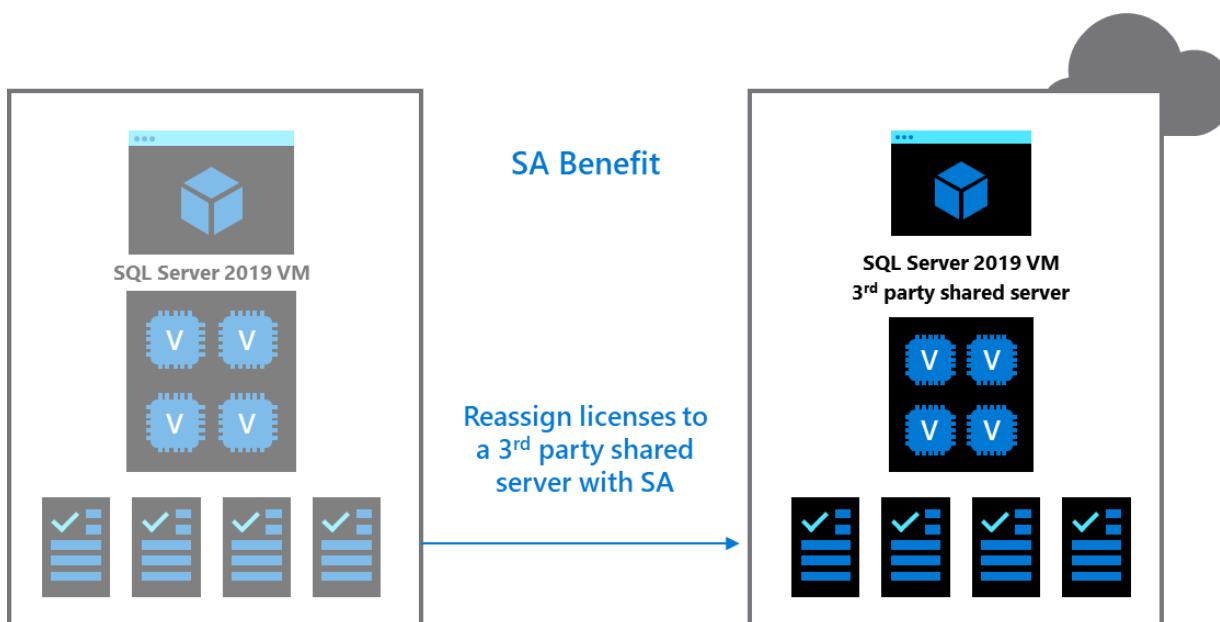


In this figure, core licenses are being reassigned within the same server farm through License Mobility.

License Mobility through Software Assurance to shared servers

Under License Mobility Through Software Assurance (SA), Customer may move SQL Server licenses covered with SA to shared servers. To use License Mobility through SA, Customer must, Run its licensed software and manage its OSEs on shared servers under the terms of its volume licensing agreement;

- Deploy its Licenses only with Microsoft Azure Services or qualified [License Mobility through Software Assurance Partner](#); and
- Complete and submit the License Mobility Validation form with each [License Mobility through Software Assurance Partner](#) who will run its licensed software on their shared servers.



In this figure, core licenses are reassigned to a 3rd party shared server through License Mobility. Note that unlike License Mobility within a server farm, licenses can only be reassigned to 3rd party only once every 90 days.

- For more information on how to use License Mobility to extend the value of SQL Servers licenses, visit <http://www.microsoft.com/licensing/software-assurance/license-mobility.aspx>

Using the Flexible Virtualization Benefit to move to the cloud

Under the Flexible Virtualization Benefit, customers may move SQL Server subscription licenses or licenses covered with SA to any Authorized Outsourcers' servers (shared or dedicated). An Authorized Outsourcer is a cloud provider that is not a "Listed Provider." The same terms and conditions that govern use on a customer's own servers apply to use of SQL Server on an Authorized Outsourcer's servers when deployed under the Flexible Virtualization Benefit. This means a customer may deploy eligible core licenses based on physical cores (i.e., fully license the server) or virtual cores (i.e., assign licenses sufficient to cover the virtual cores in each VM (subject to the four-core license minimum)). When deploying on servers that are not dedicated to its use, a customer will likely choose to license by VM.

- For more information on how to use the Flexible Virtualization Benefit to extend the value of SQL Server licenses, visit <https://aka.ms/FlexibleVirtualizationBenefitGuide>.

Licensing SQL Server for non-production use

Customers are required to license every Microsoft software product they install, configure, and use, including all physical and virtual instances. As such, licensing a development and test environment can be expensive and challenging to manage as new servers are set up and others are torn down. Microsoft offers multiple, cost-effective options for licensing SQL Server software for use in non-production environments.

SQL Server Developer Edition

SQL Server Developer Edition is a fully featured version of SQL Server software—including all the features and capabilities of Enterprise Edition—licensed for development, test and demonstration purposes only. SQL Server Developer Edition may not be used in a production environment. Any test data that was used for design, development or test purposes must be removed prior to deploying the software for production use.

Customers may install and run the SQL Server Developer Edition software on any number of devices. This is significant because it allows customers to run the software on multiple devices (for testing purposes, for example) without having to license each non-production server system.

Note: A production environment is defined as an environment that is accessed by end-users of an application (such as an internet website) and that is used for more than gathering feedback or acceptance testing of that application. Other scenarios that constitute production environments include:

- Environments that connect to a production database
- Environments that support disaster-recovery or backup for a production environment
- Environments that are used for production at least some of the time, such as a server that is rotated into production during peak periods of activity

It is rare that someone whose primary role is designing, developing, or testing software would also qualify as an “end user” of the software.

Note: Effective April 1, 2016, SQL Server Developer Edition became a free product, available through the Microsoft Dev Essentials program or as software download from the SQL Server website. For customers who need prior versions and/or additional SQL Server editions for development, test and demonstration purpose (e.g. Standard or Enterprise editions), these can be accessed through Visual Studio subscriptions.

- For more information on free Dev Essentials program, including how to register, visit <https://www.visualstudio.com/en-us/products/visual-studio-dev-essentials-vs.aspx>
- To learn more about SQL Server Developer Edition, including download, visit <https://www.microsoft.com/en-us/sql-server/sql-server-downloads>

Visual Studio subscriptions

Customers can also choose to license SQL Server software for non-production use through certain Visual Studio subscription offerings, including the Visual Studio Professional and Enterprise subscription levels. Visual Studio subscriptions are licensed on a per user basis and the software cannot be used in a production environment.

- For more information on Visual Studio subscriptions that include access to SQL Server software, visit <https://www.visualstudio.com/subscriptions/>
- For more information on Visual Studio licensing scenarios, download the Visual Studio Licensing White Paper at <https://www.visualstudio.com/wp-content/uploads/2019/05/Visual-Studio-2019-Licensing-Whitepaper-March-2019.pdf>

Product evaluations

SQL Server Evaluation Edition is a fully functional trial version of SQL Server software that automatically expires after 180 days. Microsoft Volume Licensing customers can also install and evaluate non-expiring software versions of any of the SQL Server products for 60 days before requiring a purchase.

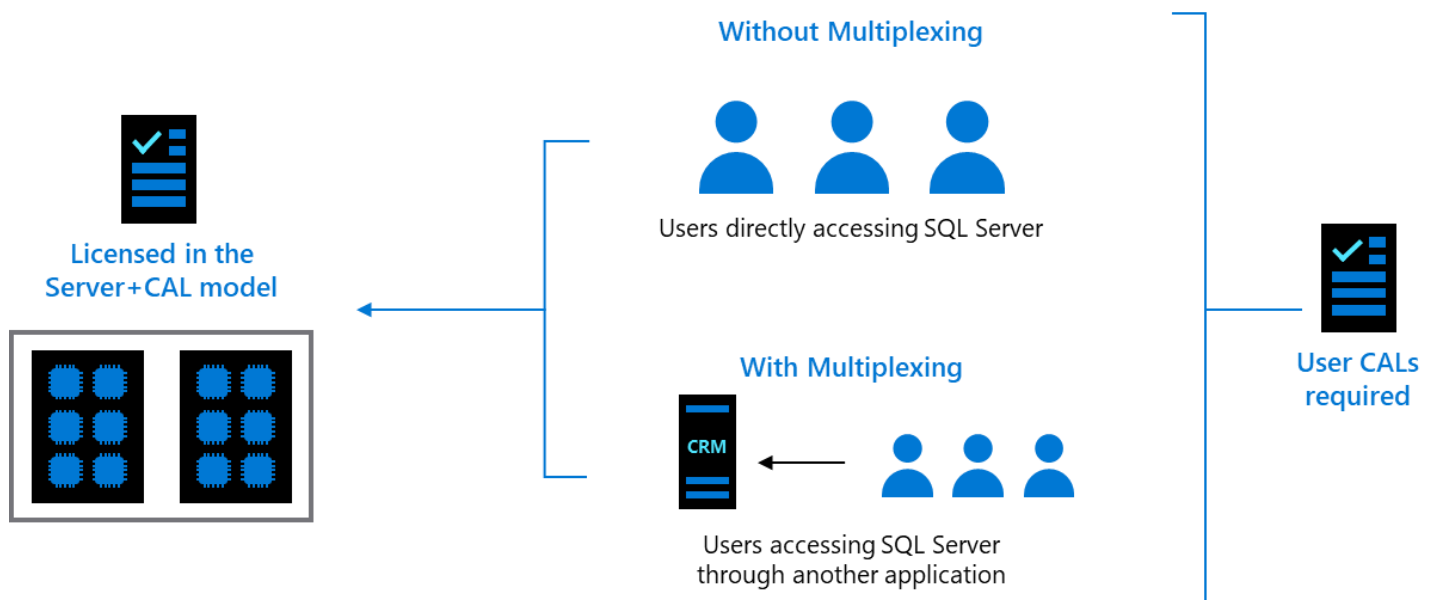
- To discover the power of SQL Server, download the free SQL Server Evaluation at <https://aka.ms/sql22download>

Licensing SQL Server in a multiplexed application environment

“Multiplexing” refers to the use of hardware or software to pool connections, reroute information, or reduce the number of devices or users that directly access or use SQL Server. Multiplexing can also include reducing the number of devices or users SQL Server directly manages.

When licensing SQL Server software under the Server+CAL licensing model, users and devices that indirectly access SQL Server data through another application or hardware device still require SQL Server CALs.

- Multiplexing does not reduce the number of Microsoft licenses required. Users are required to have the appropriate licenses, regardless of their direct or indirect connection to SQL Server.
- Any user or device that accesses the server, files, data or content provided by the server that is made available through an automated process requires a SQL Server CAL.
- The number of tiers of hardware or software between the SQL Server and the user or devices that ultimately use its data, services, or functionality does not affect the number of CALs required.
- Manual transfer of data from employee to employee does not necessitate the requirement of a CAL for the receiving employee. For example, if an employee sends a Microsoft Office Excel® version of a report to another employee, the receiving employee does not require a CAL (as long as the report does not access a server running SQL Server in some way).



This figure illustrates the licenses used in the Server+CAL licensing model via multiplexing.

SQL Server CALs are required for users or devices that directly input into, query, or view data from a SQL Server database. Similarly, SQL Server CALs are required for users or devices that input data into, query, or view data from a SQL Server database through a pooling device (such as the CRM Server in the figure above). This includes users who view data through web-based applications or enter information into a database through an intermediary product.

When users actively send SQL Server data by email or other messaging technology, recipient users do not require a SQL Server CAL. With multiplexing, these rules do not change. Likewise, the paper distribution of data does not require SQL Server CALs for the recipients of the paper report. Users who receive data directly or indirectly from SQL Server require CALs, but if these users print the data, recipient users do not require a SQL Server CAL.

- For more details on how multiplexing affects the licensing of SQL Server products under the Server+CAL model, download the Multiplexing – CAL Requirements Volume Licensing Brief at <http://download.microsoft.com/download/8/7/3/8733d036-92b0-4cb8-8912-3b6ab966b8b2/multiplexing.pdf>

Additional product information

Upgrades, downgrades and step-ups

When licensing SQL Server software, several deployment options are available to support a variety of customer upgrade scenarios.

- **Version upgrade rights** are offered with subscription licenses and as a Software Assurance (SA) benefit for qualified licenses and allow customers access to upgrade their deployments at no additional cost. Existing SQL Server subscription licenses or licenses covered by SA are automatically upgraded to licenses for the latest version of the corresponding SQL Server edition.
- **Cross edition rights** are currently available for certain SQL Server products only and allow customers to deploy an alternate (usually lower) edition in place of the currently licensed edition. SQL Server cross edition rights can be combined with the version downgrade rights (available for all products offered under a Volume Licensing Agreement) that allow customers to deploy prior versions of the software in place of the currently licensed version. In some cases, rights to deploy prior versions of product editions other than the edition currently licensed may also be allowed.

Note: When using version downgrade or cross edition deployment rights, the product use rights for the originally licensed version and edition still apply.

Software deployment options for SQL Server 2022		
Customers licensed for:	Can choose to deploy:	
	Software edition	Software version
SQL Server 2022 Standard Edition Server	SQL Server Standard Server	2022 or earlier
	SQL Server Workgroup	2008 R2 or earlier
	SQL Server for Small Business	2008 R2 or earlier
SQL Server 2022 Standard Edition Core	SQL Server Standard Core	2022 or earlier
	SQL Server Web (non-SPLA only)	2008 R2 or earlier
	SQL Server Workgroup	2008 R2 or earlier
SQL Server 2022 Enterprise Edition Core	SQL Server Enterprise Core	2022 or earlier
	SQL Server Business Intelligence	2014 or earlier
	SQL Server Standard Core	2022 or earlier
	SQL Server Datacenter	2008 R2 or earlier

This table shows deployment options available to customers with SQL Server 2022 licenses. SQL Server 2022 product terms apply.

- **Edition step-ups** are offered as a Software Assurance (SA) benefit in certain Volume Licensing programs only and allow customers to move from a lower product edition. SQL Server Standard Edition Core licenses can step-up to SQL Server Enterprise Edition Core licenses only. To be eligible to step-up to a higher edition, the lower edition license must be covered by SA. Step-Ups between licensing models are not allowed.

SQL Server migration options for Software Assurance customers

To facilitate a smooth transition to the product edition and licensing model changes, Microsoft continues to offer upgrade options to help customers who have invested in Software Assurance benefits to protect their current software investments.

For SQL Server Enterprise Edition licenses with Software Assurance

Existing SQL Server Enterprise Edition server licenses continue to have tremendous value and with the availability of ongoing SA coverage, customers licensed under the Server+CAL model can retain access to the latest product enhancements and advanced capabilities of Enterprise Edition. As such, there are no programmatic conversions to core licenses. However, no new SQL Server Enterprises edition server licenses can be purchased.

Additional considerations when licensing SQL Server Enterprise under the Server+CAL model:

Server Enterprise Edition software licensed under the Server+CAL model is intended and physically limited to only run on servers with a total of twenty cores or less:

- There are two versions of SQL Server Enterprise Edition software: a server-based version and a core-based version. Customers must run the software version for which they are licensed.
- For customers running SQL Server Enterprise Edition server-based software instances in a physical environment, OSE is only permitted to access a maximum of twenty physical cores. A per instance technical limit is also enforced.
- For customers running SQL Server Enterprise Edition server licenses in virtual environments, each set of VMs associated with a single server license (up to four per server license) can only access up to twenty hardware threads of combined power at any time.

Additional product licensing resources

For more information about licensing SQL Server, including what is new with the latest version, please visit the following websites:

- For detailed SQL Server product information, including new version features, edition comparisons, benchmarks, competitive comparisons and more, visit <https://www.microsoft.com/en-us/sql-server/sql-server-2022>
- To learn more about how to buy SQL Server, including frequently asked licensing questions, view the SQL Server Buyer's Guide located at <https://www.microsoft.com/en-us/Licensing/product-licensing/sql-server.aspx>
- For SQL Server product terms, licensing briefs and other information on Microsoft Volume Licensing topics, visit <http://www.microsoftvolumelicensing.com/>